Firm Dynamics, Monopsony, and Aggregate Productivity Differences*

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Abstract

This paper studies how labor market power affects firm dynamics and aggregate productivity. We build a dynamic model of neoclassical monopsony with occupational choice, firm growth, and productivity-enhancing technology adoption. Labor market power lowers efficiency and leads to aggregate output losses by distorting the allocation of labor, entrepreneurship, and innovation decisions. The model is consistent with cross-country evidence of higher life cycle firm growth and higher productivity investment in more competitive labor markets and can explain at least 19 percent of the differences in income per capita across countries. We find that about 60 percent of the losses are attributable to a distorted selection of entrepreneurs and a lack of innovation, suggesting that efficiency losses may be greater than those estimated by previous studies.

Keywords: Monopsony, firm dynamics, technology adoption, productivity
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1 Introduction

Firms are larger and grow faster over the life cycle in high-income countries. A common explanation is that better-functioning labor markets can improve allocative efficiency, favoring the adoption of better technology (Hsieh and Klenow, 2014). Still, despite being central in the allocation of resources across firms, labor markets around the world are far from competitive and employers operate under imperfect competition almost everywhere (Amodio et al., 2024). Labor market power is a source of inefficiency that can lower returns to adopting more efficient technologies, slow down business dynamism, and explain differences in income across countries.

In this paper, we study how important labor market power is for understanding differences in firm dynamics and aggregate productivity across countries. To this end, we build a general equilibrium model of the labor market featuring an occupational choice between entrepreneurship and wage employment, dynamic investment decisions, and taste for employers à la Card et al. (2018), which limit the firm-level elasticity of labor supply to wages. In the model, agents differ in entrepreneurial productivity and the amenities they could provide as employers. Every period they choose whether to leverage their productivity to open a business or to work for wages. Because workers value employer-specific amenities, entrepreneurs hold wage-setting power and can attract employees despite paying less than their marginal product. Entrepreneurs can also invest to improve their expected future productivity subject to a fixed cost. Through employer turnover, technology adoption, and labor supply decisions, the model generates a host of facts on firm dynamics that can be compared to the data across countries.

We calibrate the model using firm-level micro-data for the Netherlands, a country characterized by a high GDP per capita, i.e., 54,000 USD (in 2017 constant prices) in the year 2020, and an estimate for the average wage markdown as low as 1.301. This value— obtained as the average firm-level ratio between the estimated marginal revenue product of labor and the wage paid (Amodio et al., 2025b)—lies at the lower margin of the estimated spectrum (Sokolova and Sorensen, 2021), and corresponds to a firm-level elasticity of labor supply of

3.318. The benchmark model successfully replicates the observed right-skewed firm size, including the concentration of employers at the top of the distribution. It also matches the average firm size growth over the life cycle, the employer turnover, the firm age distributions, and the share of firms adopting new technologies.

We then move to our counterfactual experiment. Keeping everything else constant, we progressively change the degree of competition between firms by reducing the firms-level labor supply elasticity so to mimic the estimated patterns of wage markdown across middle- and high-income countries. There are several reasons why the labor supply to a firm might be less elastic in poorer countries. Labor markets are often more fragmented due to inadequate transportation and communication infrastructure (Brooks et al., 2021a). Labor mobility costs are higher in poorer countries (Hollweg et al., 2014) and workers are less likely to live and work in urban areas (Ananian and Dellaferrera, 2024), where labor markets tend to be more competitive due to agglomeration effects (Manning, 2010; Luccioletti, 2022). Lastly, imperfect information, social ties, and institutional irregularity could also explain why the labor supply elasticity to firms increases with development (Amodio et al., 2024; Armangué-Jubert et al., 2025; Breza and Kaur, 2025).

The model reproduces the observed cross-country differences in firm growth and technology adoption. Specifically, a counterfactual reduction in the labor supply elasticity to a firm lowers the average firm size and the average firm growth, generates higher labor turnover, and lowers the share of firms investing to improve their productivity. Moreover, we show that differences in wage markdown alone can account for 27 percent of the observed variation in GDP per capita across middle- and high-income countries, and no less than 19 percent over multiple robustness checks.

In the model, labor market power slows firm dynamics and reduces aggregate income through three channels. The first channel is standard in models of neoclassical monopsony (Card et al., 2018; Dustmann et al., 2022) and it operates through the *static allocation* of workers: lower competition increases the marginal factor cost only for a subset of firms with sufficiently high productivity, spurring employment reallocation towards less-productive, lower-paying employers. The other two mechanisms are relatively novel. Under imperfect competition in the labor market, both *selection into entrepreneurship* and *technology adoption decisions* are altered: lack of competition makes amenities more important determinants of profits, allowing low-productivity agents to reap high benefits from entrepreneurship, and lowering the returns from adopting productivity-enhancing technologies. By penalizing high-productivity employers, labor market power acts as a *skill-biased* force in the labor market, similar to what has been shown for a wide array of size-dependent policies (Guner et al., 2008; Gourio and Roys, 2014; Garicano et al., 2016; Ando, 2021): both mechanisms keep firms inefficiently small and unproductive, reducing firm growth and aggregate output.

Using our calibrated model, we quantitatively decompose each channel. To do so, we compare our benchmark economy with a counterfactual economy featuring the same degree of labor market power observed in Greece. The choice of Greece reflects the following two considerations. First, Greece has approximately one-half the GDP per capita of the Netherlands (i.e., 29,000 USD, in 2017 constant prices, in the year 2020); and second, the degree of labor market competition is much weaker in Greece than in the Netherlands: the estimated wage markdown is equal to 2.623, corresponding to an elasticity of labor supply of 0.616. We find that at least 60 percent of the model-implied difference in GDP per capita between these two countries is attributable to distorted entrepreneurial decisions and lack of innovation induced by weaker labor market competition.

This paper relates to recent work on the aggregate costs of labor market power. Berger et al. (2022) find that eliminating labor market power in the US would improve allocative efficiency of labor and rise the average wage by 48 percent, contributing to increase welfare by about 6 percent of lifetime consumption. Armangué-Jubert et al. (2025) find that labor market power can explain 15 percent of the difference in GDP per capita over the development ladder. Deb et al. (2022) show that a less competitive market structure lowered the average wage of low- and high-skilled workers in the US by 12 and 11 percent, respectively. Amodio et al. (2025b) find that fully eliminating labor market power in Peru would increase earnings on average by 26 percent. Bachmann et al. (2022) show that monopsony leads firms to stay inefficiently small and invest less in marketing, and caused a 10 percent loss in aggregate productivity in East Germany. The contribution of this paper is to show that the lack of competition in the labor market also alters selection into entrepreneurship and reduces firms' incentives to innovate, resulting in larger aggregate productivity losses than previously estimated.

Our paper also belongs to the macro literature that focuses on how differences in frictions and distortions could generate the observed cross-country differences in income per capita. (e.g. Bento and Restuccia, 2017; Guner et al., 2018; Guner and Ruggieri, 2022; Da-Rocha et al., 2023; Tamkoç and Ventura, 2024). We contribute to this literature by showing that labor market competition lowers GDP per capita by reducing complementary investments in productivity-enhancing technology.

The remainder of the paper goes as follows. In Section 2 we present cross-country evidence on firm dynamics, technology adoption, and competition in the labor market. We introduce our model in Section 3. In Section 4 we describe the calibration strategy while in Sections 5 and 6 we discuss counterfactual experiments and model mechanisms. We conclude in Section 7.

2 Stylized facts

In this section, we study how firm dynamics and local labor market competition vary across countries with different incomes per capita. For this purpose, we use the World Bank Enterprise Surveys (WBES), conducted by the World Bank. WBES is an establishment-level survey, and it is a representative sample of non-agricultural and non-financial private firms with at least 5 full-time permanent employees, spanning more than 90 countries from 2006 to 2021. The survey follows a stratified sampling methodology along sectors, establishment size, and location, with a common questionnaire for more than 90 countries from 2006 to 2021.

The data covers information on firm-level sales, number of workers, labor cost, the value of machinery, cost of raw materials, and intermediate goods employed in production, together with a large set of additional plant-level demographic characteristics, e.g., age, sector, and location, among others. We complement this data with other aggregate variables, such as real GDP per capita (expressed in USD at 2017 constant prices) from the World Bank's World Development Indicators (WDI).

We restrict our focus to countries that ever had a GDP per capita of above 25,000 USD during the years in which the survey was conducted (Tamkoç and Ventura, 2024) and only consider firms with non-missing observations on annual sales, number of workers, material and capital expenditure.[1] As a result, our sample includes 37,096 firm-year observations spread over 31 countries, consisting of middle- and high-income countries. The poorest country in the sample is Kazakhstan, with a GDP per capita of 19,615 USD in 2009, while the richest one is Ireland, with a GDP per capita of 91,791 USD in 2020. Table A.1 in Appendix A.1 reports the list of countries and years included in the sample.

As it is common in the literature, we conduct our analysis at the local labor market level. In what follows, we define a local labor market as a location-industry pair, where locations are the first administrative level of the country and industries are 2-digit ISIC v.3.1.

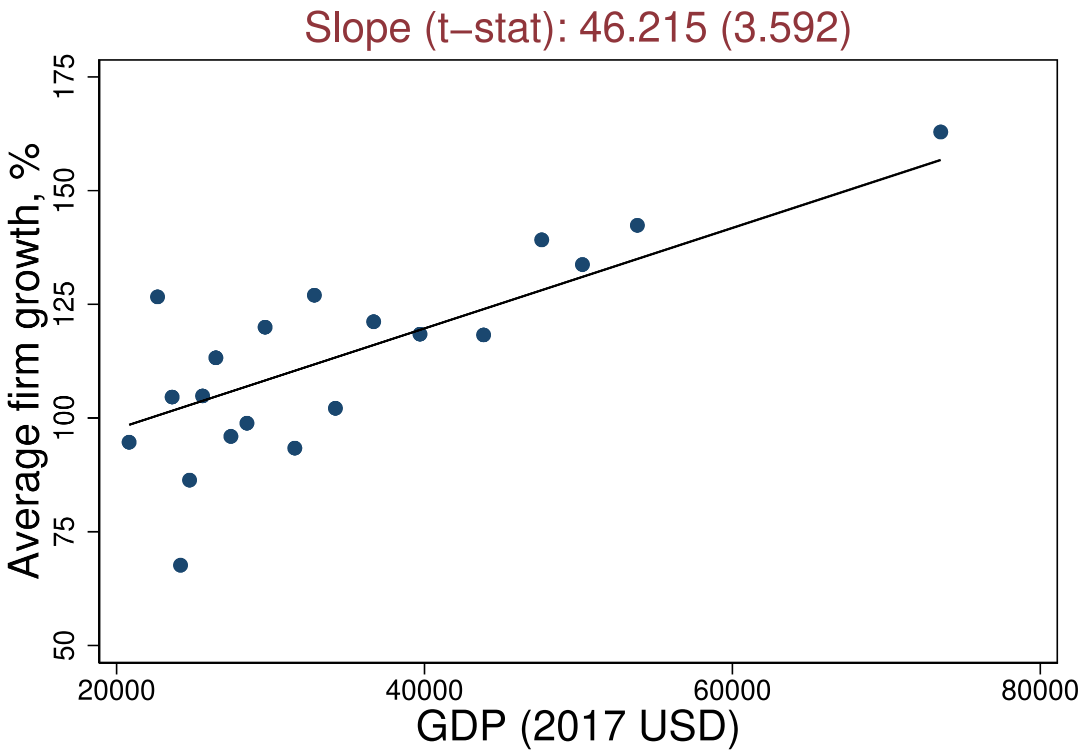
Firm dynamics. Figure 1 reports the average cumulative firm size growth across countries. Countries are ranked by their real GDP per capita. Each dot refers to the average local labor market in a country. Cumulative firm size growth is computed as a log difference between the current size and the initial size, where the former is defined as the number of employees recorded in the fiscal year before the year of the survey, whereas the latter is defined as the number of employees recorded in the first year of operations.

Firms in countries with high GDP per capita grow faster in size over their life cycle.[2] The cumulative firm size growth is about 90 percent on average in coun-

[1] Information on production input expenditure is missing for many firms in the construction and service sectors. Therefore, to ensure comparability across countries, we only focus on the manufacturing sector. See also Eslava et al. (2024) for a similar sample selection.

[2] Firms in richer countries grow faster along their life cycle even conditional on survival.

Figure 1: Average cumulative firm growth across countries



NOTES: This figure is a binscatter plot of the average cumulative firm size growth since birth across countries over GDP per capita. Fitted lines are obtained from an auxiliary regression on GDP per capita. In the caption, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

tries with a GDP per capita of 25,000 USD, and it increases to around 120 percent in countries with a GDP per capita of 60,000 USD.

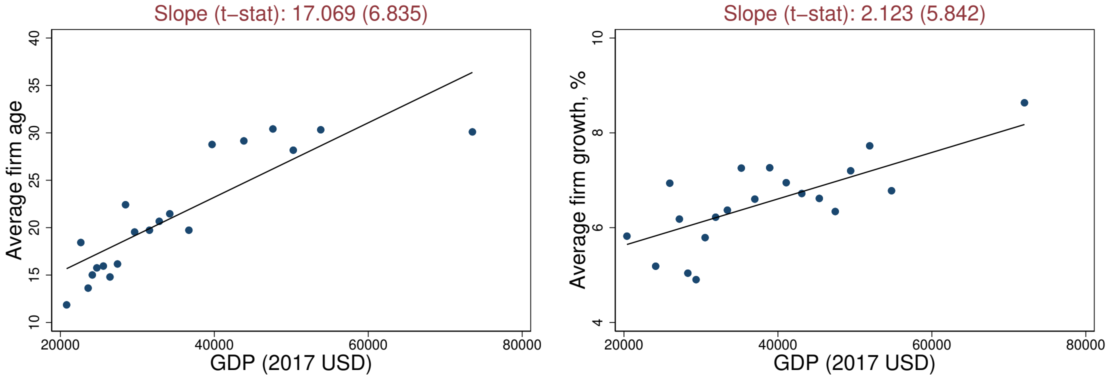
Figure 2 breaks the cumulative firm size growth into two components: firm age (panel A) versus the annualized firm size growth, residualized from firm age fixed effects (panel B). Firm age is computed by subtracting the first year of operations from the year of the survey. We compute the annualized firm size growth by dividing the cumulative firm size growth by the firm age.

Firms in countries with high GDP per capita are older on average. As we move

See Appendix A.2.

Figure 2: Firm dynamics across countries

(a) Average firm age **(b)** Average annualized firm size growth



NOTES: Panel A is a binscatter of the average firm age across countries over GDP per capita. Panel B is a binscatter plot of the average annualized firm size growth since birth across countries over GDP per capita, residualized by firm age. Fitted lines are obtained from an auxiliary regression on GDP per capita. In the caption of each panel, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

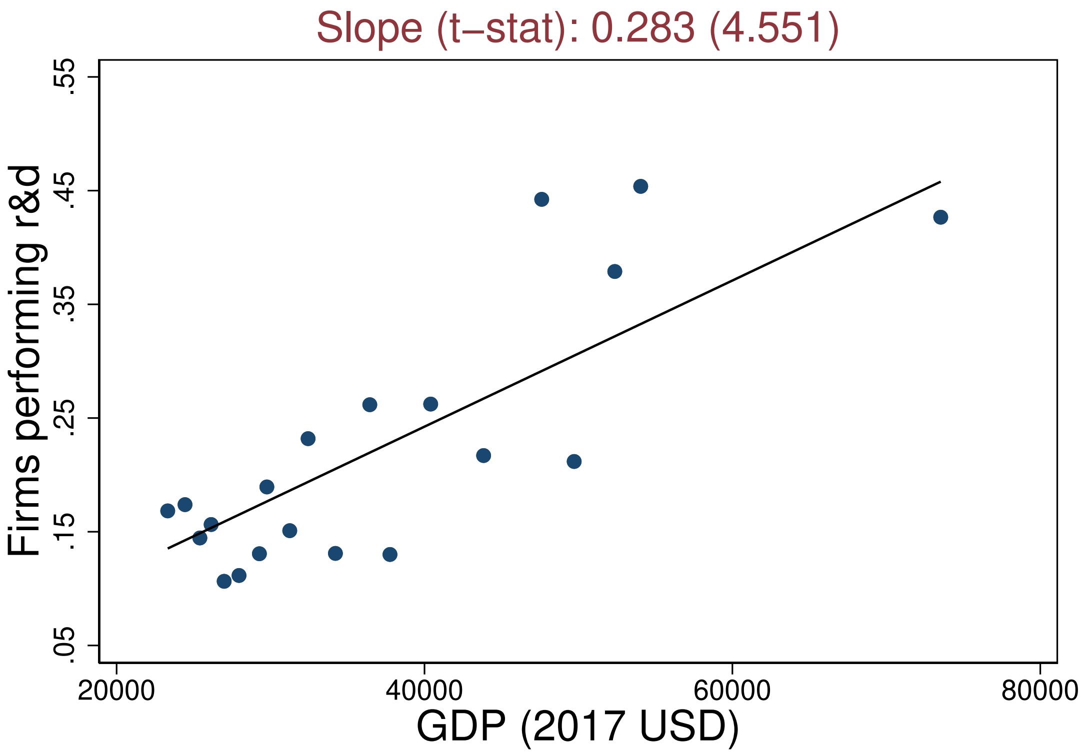
from poorer to richer countries, the average firm age triples from 11 to almost 30 years. At the same time, the annualized firm size growth increases with GDP per capita. The estimated slope implies a 1.6 percentage point higher firm growth rate per year as we move to a country with double the GDP per capita. Both relationships are significant at a 95 percent confidence level.

These facts were first uncovered by Hsieh and Klenow (2014), who showed that employment growth from birth at a firm in the US is several-fold the one observed in Mexico or India, and were replicated by Eslava et al. (2019) for the case of Colombia versus the US. Our findings are also consistent with Bartelsman et al. (2004), who document that firm churning is higher in developing countries, resulting in a higher share of new firms created over the total, and lower average firm age.[3]

Technology adoption. Figure 3 reports the share of firms that innovate across countries. Like before, each dot refers to the average local labor market in a

[3] Aga and Francis (2017) document that the probability of firm exit declines with GDP per capita. However, this relation loses significance in more conservative estimations.

Figure 3: Technology adoption across countries



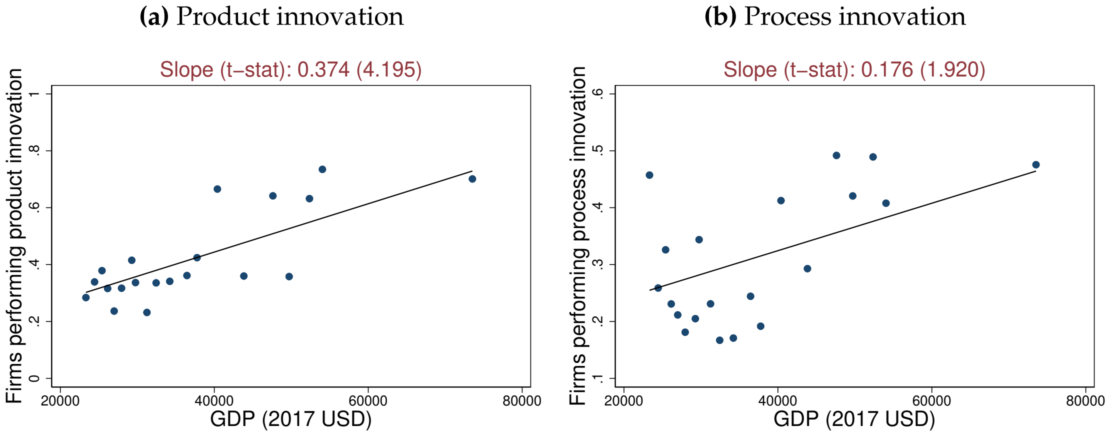
NOTES: This figure shows a binscatter of the share of firms that innovate across countries over their GDP per capita. Fitted lines are obtained from an auxiliary regression on GDP per capita. In the caption, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

country. To measure innovation, we use the share of firms that report having conducted formal research and development activities. The WBES reports answers to the question: "During last fiscal year, did this establishment spend on formal research and development activities, either in-house or contracted with other companies, excluding market research surveys?" We construct a binary variable for firm-level innovation by assigning a value of 1 if a firm reports a positive spending on formal research and development activities, 0 otherwise.

Firms in high GDP per capita countries are more likely to perform R&D, whether in-house or contracted. As we move from poorer to richer countries, the share of firms that spend on R&D more than doubles, from around 15 percent to more than 40 percent.

Figure 4 complements Figure 3 by reporting the share of firms performing product innovation (panel A) and process innovation (panel B) across countries with different GDP per capita. These two variables are constructed using answers collected by the WBES to the following two questions: i) "During Last 3 Yrs, Establishment Introduced New/Significantly Improved Process?"; and ii) "New Products/Services Was Introduced Over Last 3 Years?". We construct binary variables for product and process innovation assigning a value 1 to a firm answers affirmatively to the either question, 0 otherwise.

Figure 4: Firm innovation over development



NOTES: Panel A binscatters the share of firms that conduct product innovation across countries over their GDP per capita. Panel B binscatters the share of firms that conduct process innovation across countries over their GDP per capita. Fitted lines are obtained from auxiliary regressions on GDP per capita. GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

Firms in richer countries are also more likely to conduct both product and process innovation: as we move from middle to high-GDP per capita countries, the share of firms performing product innovation increases from 20 to 80 percent. Similarly, the share of firms performing process innovation increases from 20 to 50 percent.[4]

This evidence is consistent with Lederman and Maloney (2003), who document

[4]These facts are robust to alternative data sources. See Appendix A.3 for a discussion.

that R&D expenditure (measured as a share of GDP) is higher in richer countries, and it complements recent findings of Farrokhi et al. (2024), who document relatively low adoption of new technology in low-income countries.

Labor market competition. Finally, we compare the degree of local labor market competition across countries and use firm-level wage markdown as a proxy for labor market power. Specifically, we construct wage markdowns, μ_{it} for firm i at time t as a ratio between the firm-level marginal revenue product of labor and the wage paid (Brooks et al., 2021b; Yeh et al., 2022; Pham, 2023), i.e.

$$\mu_{it} = \frac{\text{MRPL}_{it}}{w_{it}}.$$

Denoting the revenue elasticity of labor as ξ, we can express the wage markdown as a ratio between the revenue elasticity of labor and the wage-bill share of total revenue, i.e.

$$\mu_{it} = \frac{\xi}{\frac{w_{it}\ell_{it}}{y_{it}}}.$$

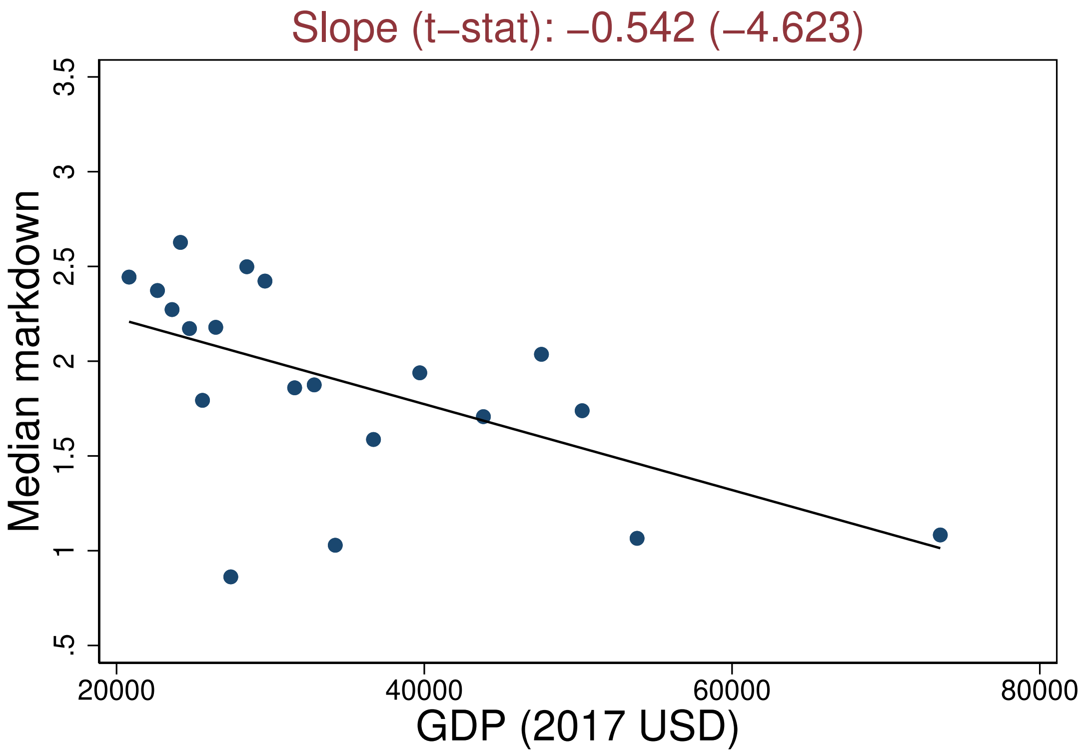
We measure average revenues, y_{it}/ℓ_{it}, and average wage, w_{it}, using firm annual sales per number of employees and annual payroll per number of employees, respectively, while we estimate ξ separately for each country in the sample using a standard control function approach. We report details of the estimation procedure in Appendix A.4.

Figure 5 scatters the median firm-level markdown in the average local labor market of the sampled countries against GDP per capita. Local labor markets are more competitive in richer countries, and firms charge lower markdowns.[5]

In countries with a GDP per capita of 25,000 USD, we estimate a median markdown of 2.25 on average; that is, workers are paid about 55 percent less than their marginal product. These values fall in the range of estimates for middle-income countries obtained from other studies in the literature (Sokolova and Sorensen, 2021). For instance, exploiting variation in firms' exposure to trade and exchange rates, Amodio et al. (2025a) trace out the firm-level labor supply

[5]This pattern is robust to estimating firm-level markdown controlling for several firm-level observables. See Figure A.3 in Appendix A.4.

Figure 5: Median markdown across countries

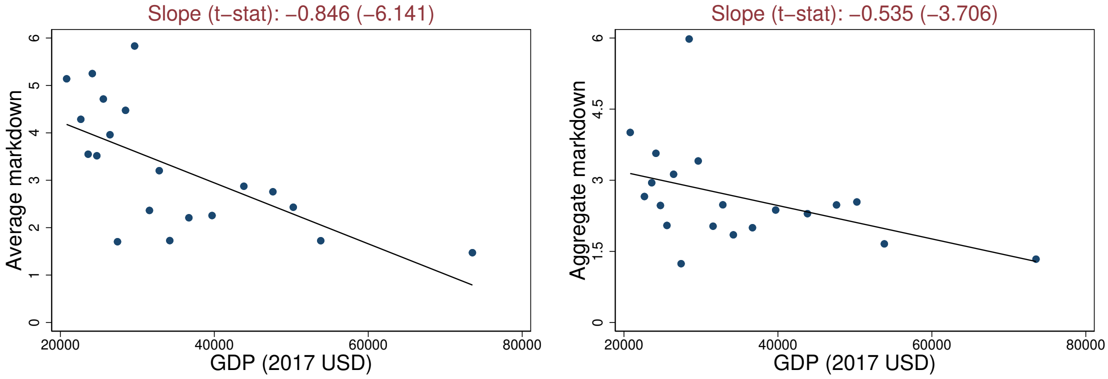


NOTES: This figure shows a binscatter of the median firm-level markdown across countries over GDP per capita. Fitted lines are obtained from an auxiliary regression on log GDP per capita. In the caption of each panel, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

curves of several Latin American countries (GDP per capita ranging from 2,000 USD in Nicaragua to 17,000 USD in Chile, in 2022), and estimate an average labor supply elasticity of 0.82, corresponding to a wage markdown of 2.2. Bassier (2023) uses matched employer-employed data for South Africa (GDP per capita of 6,000 USD in 2022) and obtains an estimate for the firm labor supply elasticity of 0.74, corresponding to a wage markdown of 2.35. Garcia-Louzao and Ruggieri (2023) exploits wage variation for workers moving across firms and estimate a labor supply elasticity for Lithuania (GDP per capita of 23,065 USD in 2013) ranging between 0.7 and 0.9, which corresponds to wage markdowns between 2.1 and 2.4. Ogloblin and Brock (2005) document that male workers

Figure 6: Mean vs. aggregate markdown across countries

(a) Unweighted average markdown　　**(b)** Aggregate (sales-weighted) markdown



NOTES: Panel A shows a binscatter of the unweighted average markdown across countries over GDP per capita. Panel B shows the aggregate (sales-weighted) markdown across countries over GDP per capita. Fitted lines are obtained from an auxiliary regression on GDP per capita. In the caption of each panel, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

in Russia (GDP per capita of 25,933 USD in 2012) earn around 35.5 percent less than their potential competitive wage, corresponding to a wage markdown of 1.55.

In high-income countries, the estimated wage markdown is much lower. The median markdown is about 1.25 in countries with a GDP per capita of 60,000 USD, which means workers are paid 20 to 25 percent less than their marginal product. The estimated wage markdowns for high-income countries fall within the range of estimates of 17 and 24 percent provided by Azar et al. (2022) and Berger et al. (2022) for the United States. It also lies between 16 and 25 percent, the estimates obtained by Datta (2022) for the United Kingdom.

Figure 6 complements Figure 5 by reporting the unweighted average markdown (panel A) and the aggregate (sales-weighted) markdown (panel B) across countries. Both measures of markdowns are larger than the median ones but follow a similar pattern over development: labor market power decreases with development regardless of how markdowns are aggregated across firms.

This evidence is consistent with Armangué-Jubert et al. (2025), who show that markdowns decrease with income per capita for countries with GDP per capita over 2,000 USD. This evidence is also consistent with Eslava et al. (2024), who document that markdowns are higher in less developed countries. Using WB-ES data, they show this pattern to be robust across different econometric specifications and alternative sample selections. Their estimated markdown decreases from values around 2 in middle-income economies to 1.2 in high-income countries. Similar to our findings, their estimates of aggregate (sales-weighted) markdowns also decline with GDP per capita, going from about 4 for middle-income countries to 2 for high-income countries. Amodio et al. (2024) document a hump-shaped relationship between GDP per capita and median markdowns for relatively poorer countries than those in our sample, i.e., those with GDP per capita levels below 25,000 USD. Our estimates of median wage markdowns for countries just above this threshold are comparable (in magnitude) to their estimates for countries just below the threshold.

To summarize, firms grow faster and are more likely to innovate in richer countries, where local labor markets are more competitive. In the next section, we build a dynamic model of neoclassical monopsony that fits our stylized facts and use it to study how labor market power affects firm dynamics, productivity-enhancing technology adoption, and aggregate productivity.

3 Model

We extend a standard model of neoclassical monopsony, as discussed in Card et al. (2018) and Dustmann et al. (2022), to a dynamic general equilibrium setting with an entrepreneurial choice and endogenous productivity investment.

Time is discrete. The economy is populated by a unitary measure of agents, each characterized by entrepreneurial productivity z defined over a ladder $\mathcal{Z} = [\underline{z}, ..., z_{-}, z, z_{+}, ..., \overline{z}]$, and amenity a defined over a subset $\mathcal{A}$ of the reals. Agents face a stochastic life cycle, discount the future using a discount factor β, and have a probability of exiting the labor market equal to δ_w. Exiting agents are replaced by an equal share of new entrants who draw a tuple of characteristics (z, a) from two independent distributions, $\Psi_z(z)$, and $\Psi_a(a)$. In what follows we assume both distributions to be log-normal with mean 0 and variances σ_z^2 and σ_a^2, respectively.[6]

Each period following entry, agents decide whether to become wage workers or entrepreneurs. Let L and $E = 1 - L$ denote the (endogenous) aggregate measures of workers and entrepreneurs in the economy, respectively. Entrepreneurial productivity of every agent evolves stochastically over the life cycle, following a discrete time Poisson process which moves it one step up or down the productivity ladder with probability p_n and 1-p_n (Shimer, 2005). Entrepreneurs can invest in innovation, which increases the likelihood of moving up the ladder to $p_i > p_n$, resulting in a higher expected future productivity.

Both workers and entrepreneurs are hand to mouth, they value consumption as well as workplace amenities. Entrepreneurs produce a homogeneous final good, whose price is the numeraire of the economy. Finally, labor markets are assumed to be spot markets that clear every period: entrepreneurs post wages to maximize their profits, with knowledge of workers' labor supply function. Workers observe posted wages and amenities and choose which firms to work for. Job differentiation through amenities endows entrepreneurs with wage-setting power.

3.1 The problem of the workers

Workers derive utility from consumption of a final good and from the work environment that a firm provides. The instantaneous indirect utility for a worker i employed by entrepreneur (firm) j is:

$$u(z_i, a_i, z_j, a_j, \nu_{ij}) = \epsilon^L \ln(w_j) + a_j + \nu_{ij}, \tag{1}$$

with $\epsilon^L > 0$. In equation (1), w_j is the wage paid by firm/entrepreneur j; a_j denotes the amenity provided by firm j, common to all workers within the firm, while ν_{ij} is a match-specific i.i.d. preference shock for working for firm j, assumed to follow a Gumbel distribution with location parameter 0 and scale

[6] Assuming independent distributions allows us to restrict the parametric space and simplifies the identification strategy. We will relax this assumption in Section 5.2.

parameter σ_ν.

The specification of preferences allows for *vertical* and *horizontal* employer differentiation. Both amenities and idiosyncratic preference shocks are meant to capture non-pecuniary job characteristics, such as schedule flexibility, commuting arrangements, workplace safety, provision of parental leave, job autonomy and security, among others (Maestas et al., 2023; Mas and Pallais, 2017; Sorkin, 2018; Sockin, 2024). On the other hand, while differences in a_j imply a ranking across employers, the term v_{ij} makes workers heterogeneous in their preferences over the same firm.

The indirect utility function in equation (1) is also convenient because it offers a direct map between the parameter ϵ^L and the elasticity of labor supply: lower values of ϵ^L increases the relative utility weight of amenities, making workers' individual labor supply to a firm less responsive to wage changes. It also offers a micro-foundation of the CES preferences for differentiated jobs used, among others, by Berger et al. (2022) and Felix (2022).[7]

Let $\tilde{U}(z_i, a_i)$ be the expected value of being a wage worker for agent i, before a realization of ν_{ij} is observed. Because ν_{ij} is assumed to be independent across alternatives and type-I extreme value distributed, $\tilde{U}(z_i, a_i)$ can be written as

$$\begin{aligned}\tilde{U}(z_i, a_i) &= \mathbb{E}\left[\max_k \{U(z_i, a_i, z_k, a_k) + \nu_{ik}\}\right] \\ &= \sigma_\nu \ln\left(E\int_{\mathcal{Z}\times\mathcal{A}} \exp\left(\frac{U(z_i, a_i, z_k, a_k)}{\sigma_\nu}\right)\psi(z_k, a_k)dz_k da_k\right),\end{aligned}$$

where $U(z_i, a_i, z_j, a_j)$ denotes the worker-i's conditional value of working at firm j, which is equal to

$$\begin{aligned}U(z_i, a_i, z_j, a_j) &= \epsilon^L \ln(w_j) + a_j \\ &+ \beta(1-\delta_w)\big(p_n \max\{\tilde{U}(z_{i+}, a_i), V(z_{i+}, a_i)\} + (1-p_n)\max\{\tilde{U}(z_{i-}, a_i), V(z_{i-}, a_i)\}\big),\end{aligned} \tag{2}$$

[7] At the aggregate level, CES preferences generate the same labor supply system as in a framework where workers solve a discrete choice problem and firms compete in a monopsonistic labor market. This argument adapts those by Anderson et al. (1988), who establish an equivalence between single-sector CES and single-sector logit, and Verboven (1996), who extends it to nested-logit and nested-CES environments. See Berger et al. (2022), Appendix B for a formal discussion.

while $\psi(z_k, a_k)$ denotes the equilibrium distribution of active entrepreneurs across productivity and amenities, and V is the value of being an entrepreneur, defined below.

Notice that entrepreneurial productivity evolves exogenously for wage workers, i.e., it increases by one step on the ladder with probability p_n, while it decreases with the opposite probability, $1 - p_n$. Notice also that every periods agents face to option of remaining a wage worker or becoming an entrepreneur. The max operator in equation (2) implies a policy function for entrepreneurial choice, $\rho^e(z_i, a_i)$, defined as

$$\rho^e(z_i, a_i) = \begin{cases} 1 & \text{if} \quad V(z_i, a_i) > \tilde{U}(z_i, a_i), \\ 0 & \text{otherwise} \end{cases}$$

Finally, wage worker i will choose to work for entrepreneur j if

$$U(z_i, a_i, z_j, a_j) + \nu_{ij} \geq U(z_i, a_i, z_k, a_k) + \nu_{ik}, \quad \forall k \neq j.$$

Given the assumption on ν_{ik}, the probability that a wage worker i chooses to work for a firm j is given by the following continuous logit formulation (McFadden, 1976; Ben-Akiva et al., 1985):

$$p_{ij} = \frac{\exp\left(\frac{U(z_i, a_i, z_j, a_j)}{\sigma_\nu}\right)}{\int_L^1 \exp\left(\frac{U(z_i, a_i, z_k, a_k)}{\sigma_\nu}\right) dk}.$$

The aggregate labor supply to a firm j is then equal to:

$$L(z_j, a_j) = L_j = L \int_{\mathcal{Z} \times \mathcal{A}} p_{ij} \phi(z_i, a_i) dz_i da_i \tag{3}$$

where $\phi(z_i, a_i)$ is the equilibrium distribution of workers across productivity and amenities. Re-arranging terms, equation (3) can be re-written as to:

$$L_j = L\Theta \exp\left(\epsilon^L \ln(w_j) + a_j\right)$$

where Θ is an aggregate market shifter.[8] The labor supply solution resembles the one obtained in Card et al. (2018): because the labor market is a spot market, dynamic forces only affect Θ.

3.2 The problem of the entrepreneurs

Entrepreneurs produce a homogeneous final good using a decreasing return to scale production function,

$$Y_j = z_j L_j^{\xi}, \tag{4}$$

where z_j denotes their entrepreneurial productivity, L_j is the labor supplied to their firm, and $\xi \in (0, 1)$ denotes the revenue elasticity of labor. Every period, entrepreneurs post a wage w_j to maximize profits given knowledge of the labor supply function in equation (3). Since entrepreneurs do not observe the preference shocks of individual workers, they cannot perfectly discriminate and will offer the same wage to all of their workers.

The static problem of an entrepreneur j is then given by

$$\max_{w_j} \pi_j(z_j, a_j) = z_j L_j^{\xi} - w_j L_j \tag{5}$$
$$\textit{subject to} \quad L_j = L\Theta \exp\left(\epsilon^L \ln(w_j) + a_j\right).$$

where a_j denotes the amenity provided by entrepreneur j to their employees. A solution to this problem is an optimal wage schedule, $W(z_j, a_j), \forall j$.

Given the solution to the static profit maximization problem, entrepreneurs choose whether to invest in their productivity. Innovation allows entrepreneurs to increase their expected productivity by raising the likelihood of productivity improvement to $p_i > p_n$, hence, by construction, lowering the likelihood of productivity depreciation. To innovate, entrepreneurs incur a per-period fixed cost c_x, while all entrepreneurs pay a fixed cost of operation c_f. Both costs are defined in terms of final goods.[9]

[8] We report the full derivation of the labor supply function in Appendix B.1.
[9] In Section 5.3, we test the robustness of our finding to defining both costs in terms of la-

The value to an agent i of being an entrepreneur is then given by

$$V(z_i, a_i) = \max\{V^I(z_i, a_i), V^N(z_i, a_i)\} \tag{6}$$

where $V^I(z_i, a_i)$ is the value of investing in productivity, equal to

$$\begin{aligned} V^I(z_i, a_i) &= \epsilon^L \ln(\pi(z_i, a_i) - c_f - c_z) + a_i \\ &+ \beta(1-\delta_w)\big(p_i \max\{V(z_{i+}, a_i), \tilde{U}(z_{i+}, a_i)\} + (1-p_i)\max\{V(z_{i-}, a_i), \tilde{U}(z_{i-}, a_i)\}\big) \end{aligned}$$

while $V^N(z_i, a_i)$ is value of not investing,

$$\begin{aligned} V^N(z_i, a_i) &= \epsilon^L \ln(\pi(z_i, a_i) - c_f) + a_i \\ &+ \beta(1-\delta_w)\big(p_n \max\{V(z_{i+}, a_i), \tilde{U}(z_{i+}, a_i)\} + (1-p_n)\max\{V(z_{i-}, a_i), \tilde{U}(z_{i-}, a_i)\}\big) \end{aligned}$$

Notice that entrepreneurs value their own amenities, which they also provide to their employees. We interpret this as the consequence of the entrepreneurs sharing the same work environment of their employees (Stephan et al., 2024). This choice also ensures that the instantaneous indirect utility functions of entrepreneurs and wage workers are consistent.[10]

Finally, the max operator in equation (6) implies a policy function for investment into innovation, $\rho^z(z_i, a_i)$, defined as

$$\rho^z(z_i, a_i) = \begin{cases} 1 & \text{if} \quad V^I(z_i, a_i) > V^N(z_i, a_i), \\ 0 & \text{otherwise.} \end{cases}$$

3.3 Equilibrium

A stationary recursive equilibrium for this economy is a list of value functions $V(z_i, a_i)$, $U(z_i, a_i, z_j, a_j)$ and $\tilde{U}(z_i, a_i)$, an associated entrepreneurship policy function $\rho^e(z_i, a_i)$ and innovation policy function $\rho^z(z_i, a_i)$, a wage schedule $W(z_i, a_i)$, an allocation of labor $L(z_i, a_i)$, an aggregate measure of workers L,

bor.

[10]We experimented with a version of the model where entrepreneurs are precluded from valuing their own amenities and their indirect utility only depends on profit flows. We found no significant quantitative differences.

a distribution of agents over productivity and amenities, $\Omega(z_i, a_i)$, and distributions of wage workers and entrepreneurs over productivity and amenities, $\phi(z_i, a_i)$ and $\psi(z_i, a_i)$, such that:

- The labor supply to firm i, $L(z_i, a_i)$, satisfies equation (3);
- $\rho^e(z_i, a_i)$ and $\rho^z(z_i, a_i)$ solve the entrepreneurial and the innovation choices, and the value functions $V(z_i, a_i)$, $U(z_i, a_i, z_j, a_j)$ and $\tilde{U}(z_i, a_i)$ attain their maxima;
- The aggregate measure of workers is consistent with the entrepreneurial choices:

$$L = \int_{\mathcal{Z}\times\mathcal{A}} L(z_i, a_i)\psi(z_i, a_i)dz_i da_i = \int_{\mathcal{Z}\times\mathcal{A}} (1-\rho^e(z_i, a_i))\Omega(z_i, a_i)dz_i da_i;$$

- The distribution of agents over productivity and amenities, $\Omega(z_i, a_i)$ is stationary and replicates itself through entry and exit, and the policy functions, as in equations (14), (15) and (16), defined in Appendix B.2.
- The distributions of wage workers and entrepreneurs over productivity and amenities are stationary and defined as

$$\phi(z_i, a_i) = \frac{(1-\rho^e(z_i, a_i))\Omega(z_i, a_i)}{\int_{\mathcal{Z}\times\mathcal{A}} (1-\rho^e(z_i, a_i))\Omega(z_i, a_i)dz_i da_i},$$

and

$$\psi(z_i, a_i) = \frac{\rho^e(z_i, a_i)\Omega(z_i, a_i)}{\int_{\mathcal{Z}\times\mathcal{A}} \rho^e(z_i, a_i)\Omega(z_i, a_i)dz_i da_i},$$

respectively.

A solution algorithm is presented in Appendix B.3.

3.4 Discussion

In the model, competition in the labor market operates as a “skill-biased" force, in the sense of favoring high-productivity entrepreneurs, and it does so through different channels.

To gain some insights, let us assume that the aggregate labor supply L is fixed and constant. Notice that profit maximization (5) subject to equation (3) yields the following equilibrium employment choice by firm j:

$$\ln(L_j) = \frac{\epsilon^L}{1+(1-\xi)\epsilon^L}\ln(z_j) + \frac{1}{1+(1-\xi)\epsilon^L}a_j + C$$

where $C = \frac{1}{1+(1-\xi)\epsilon^L}\left[\epsilon^L \ln\left(\frac{\epsilon^L}{1+\epsilon^L}\right) + \epsilon^L \ln \xi + \ln(L) + \ln(\Theta)\right]$ is a market-level constant. Rearranging the equation above, we obtain that the relative employment between firms with a low- and high-productivity, $\underline{z}$ and $\overline{z}$, and same amenities a, equals:

$$\frac{L(\overline{z}, a)}{L(\underline{z}, a)} = \left(\frac{\overline{z}}{\underline{z}}\right)^{\frac{\epsilon^L}{1+(1-\xi)\epsilon^L}} \tag{7}$$

Similarly, the relative employment between firms with low- and high-amenities, $\underline{a}$ and $\overline{a}$, and the same level of productivity z, is equal to:

$$\frac{L(z, \overline{a})}{L(z, \underline{a})} = \left(\frac{\exp(\overline{a})}{\exp(\underline{a})}\right)^{\frac{1}{1+(1-\xi)\epsilon^L}} \tag{8}$$

Equations (7) and (8) predict that when the labor supply elasticity rises, relative employment falls at the lower-productivity and higher-amenities firms. This effect is standard in static models of classical monopsony (Card et al., 2018; Autor et al., 2023; Armangué-Jubert et al., 2025). With a constant aggregate labor supply L, an equilibrium reduction in relative employment at lower-productivity and higher-amenities firms implies labor reallocation towards high-productivity firms and away from high-amenities firms. We summarize this result in our first proposition.

Proposition 1 *Fix the aggregate labor supply. Everything else equal, labor reallocates towards high-productivity firms and away from high-amenities firms when the labor supply elasticity, ϵ^L, increases.*

Proof 1 *Immediate from equations (7) and (8). Full derivation in Appendix B.4.*

Through reallocation of employment across firm types, higher labor competition increases allocative efficiency. To the extent that resource allocation is more efficient when firms' labor market power is lower, a higher elasticity of labor supply ϵ^L will result in lower differences in the average revenue product of labor (APL_j) across firms.[11] To see this, notice that the elasticities of APL_j with respect to firm-level productivity, z_j, and amenities, $\exp(a_j)$, are equal to

$$\frac{\partial \ln \text{APL}_j}{\partial \ln z_j} = 1 - \frac{(1-\xi)\epsilon^L}{1+(1-\xi)\epsilon^L}, \tag{9}$$

and

$$\frac{\partial \ln \text{APL}_j}{\partial a_j} = -\frac{(1-\xi)}{1+(1-\xi)\epsilon^L}, \tag{10}$$

respectively. When ϵ^L is low, the average revenue product becomes largely dispersed across firms: it increases with firm-level productivity and declines with firm-level amenities. Which suggests high-productivity low-amenity firms face higher barriers in less competitive markets, rendering them smaller than optimal. As ϵ^L increases, both $\partial \ln \text{APL}_j / \partial \ln z_j$ and $\partial \ln \text{APL}_j / \partial a_j$ approach zero, and APL_j equalize across firms. We summarize this result in the following proposition.

Proposition 2 *Fix the aggregate labor supply. The dispersion of the average product of labor across firms reduces when the labor supply elasticity, ϵ^L, increases.*

Proof 2 *Immediate from equations (9) and (10). Full derivation in Appendix B.4.*

Like employment, profits reallocates from high-amenities firms to high-productivity firms when the elasticity ϵ^L increases. Notice that the elasticities of firm-j variable profits with respect to firm-level productivity, z_j, and amenities, $\exp(a_j)$,

[11] Labor misallocation generates differences in the marginal revenue products of labor (Hsieh and Klenow, 2009, 2014). Given production technology, dispersion in marginal revenue product result into dispersion of average revenue product, as the latter is proportional to the former.

are equal to

$$\frac{\partial \ln \pi_j}{\partial \ln z_j} = \frac{1+\epsilon^L}{1+(1-\xi)\epsilon^L}, \tag{11}$$

and

$$\frac{\partial \ln \pi_j}{\partial a_j} = \frac{\xi}{1+(1-\xi)\epsilon^L}, \tag{12}$$

respectively. The increase in variable profits from a proportional increase in productivity is higher when the labor supply elasticity ϵ^L is high. On the other hand, variable profits increase less with amenities when the labor supply elasticity ϵ^L is high. We summarize this result in our third proposition.

Proposition 3 *Fix the aggregate labor supply. Variable profits are more (less) correlated to firm-level productivity (amenities) when the labor supply elasticity, ϵ^L, increases.*

Proof 3 *Immediate from equations (11) and (12). Full derivation in Appendix B.4.*

With a low labor supply elasticity, agents with high amenities have wage-setting power as entrepreneurs. High enough to attract enough workers to produce and make sufficient profits to compete in the market, even with relatively low productivity. When labor supply elasticity increases, the competitive advantage shifts away from high-amenities firms and moves towards high-productivity firms. This alters the decision of becoming entrepreneurs, improving selection in favor of high-productivity agents. It also alters the decision of adopting productivity-enhancing technology, which becomes more profitable as the return from investment is increasing in the labor supply elasticity. This results in higher firm growth, higher productive efficiency, and higher output per capita.

In the next section, we use our model to quantify how much each mechanism, i.e., labor allocation across employers, selection into entrepreneurship, and innovation decision, contributes to fostering firm dynamics and productivity, allowing labor supply to react to changes in labor market power.

4 Estimation

We discipline the model using WBES data for the Netherlands, one of the richest countries in the sample, with an annual GDP per capita of 54,275 USD. We follow Armangué-Jubert et al. (2025) and calibrate the model to replicate the average labor market in the country, as defined by a region-industry pair.

Some parameters are calibrated without solving the model. We chose a model period of a year. We normalize the scale parameter of the Type-I GEV shock, σ_ν, to 1. We set the discount factor, β, to 0.961, consistent with an annual interest rate of 0.04, and choose δ_w to be 0.025 such that agents spend on average 40 years in the labor market. We calibrate the revenue elasticity of labor, ξ, using the corresponding estimate obtained in Section 2, i.e., 0.333. Finally, we use the estimated wage markdown to back out the labor supply elasticity. Given the monopsonistic labor market structure, the elasticity of labor supply is equal to

$$\epsilon^L = \frac{1}{\mu - 1}$$

where μ is the wage markdown for firms in the local labor market. We set μ equal to the median wage markdown in the Netherlands. In Section 2, we estimated this value to be 1.301. Which implies a labor supply elasticity of 3.318. Table C.1 in Appendix C.1 summarizes the value of these parameters and their targets.

The remaining parameters, collected in the following vector,

$$\vartheta := \{c_f, c_x, p_i, p_n, \sigma_z, \sigma_a\},$$

are estimated using the method of simulated moments. The estimator $\hat{\vartheta}$ is the minimizer of the following objective function:

$$\hat{\vartheta} = \arg\min_{\vartheta} d(\vartheta)' W d(\vartheta)$$

where $d(\vartheta)$ denotes the absolute distance between a vector of empirical targets, $\bar{g}$ and their model counterpart, $g(\vartheta)$, while the weighting matrix, W, is chosen to be diagonal with entries equal to the squared inverse of each empirical moment.[12]

Table 1: Parameters estimates and standard errors

Parameters	Description	Estimates	St.Error
c_f	Operating costs	22.80	0.206
c_x	Innovation costs	122.4	1.343
p_i	Productivity growth of investors	0.649	0.274
p_n	Productivity growth of non-investors	0.499	0.020
σ_z	Productivity dispersion	2.134	0.044
σ_a	Amenities dispersion	0.899	0.069

NOTES: The table shows the list of estimated parameters, their point estimates, and the standard errors computed using the Delta method.

Table 1 describes the list of estimated parameters, their point estimates, together with standard errors computed using the standard asymptotic variance expression (Newey and McFadden, 1994).[13] Table C.2 in Appendix C.1 reports the list of targeted moments and their model counterpart.[14]

The operating cost, c_f is estimated to match an average firm size of 59 employees while the innovation cost is chosen to match a share of firms investing in R&D of 41 percent. The productivity dynamics of investors, p_i, is informed by the average cumulative employment growth rate since entry among observed incumbents. In the data, the latter amounts to 153 percent. The average firm age, which is approximately 30 y.o., is instead meant to discipline the productivity dynamics of non-investors, p_n through entry and exit into entrepreneurship. Finally, the dispersions in entrepreneurial talents at entry, σ_z, and amenities, σ_a, are disciplined by the standard deviation of (log) firm size (0.988) and (log)

[12]We opted for this weighting matrix to ensure the stability of our estimator while maintaining consistency and keeping the estimation loss independent of units of measurement.

[13]Specifically, the variance covariance matrix is $(J'WJ')^{-1}J'W\hat{Q}WJ(J'WJ')^{-1}$, where J is the Jacobian matrix, whose (i,j)-entry is equal to $\partial d(\vartheta(i))/\partial\vartheta(j)$, while $\hat{Q} = \hat{\text{cov}}(d(\vartheta))$ is bootstrapped from the sample data with 1000 replications.

[14]While our estimation allows firm size to be arbitrarily small, our database does not cover plants with less than 5 workers. To avoid any inconsistency, we apply the same truncation to our simulated moments (see Coşar et al. (2016) for a similar approach). In Section 5.4, we test the robustness of our results to this approach and re-estimate the model using empirical moments imputed to cover a representative sample of of *existing* producers.

wages (0.418), respectively. The fit of the model is very satisfactory: the sum of squared deviations between empirical and simulated moments is equal to 1.7 percent.[15]

The estimates of c_f and c_x are 22.8 and 122.4, respectively. These values correspond to about 8 and 43 percent of the average profits made by incumbent firms in the model. Jump probabilities, p_i and p_n, are estimated to be 0.65 and 0.50, respectively; i.e., firms adopting productivity-enhancing technology are 15 percent more likely to grow compared to those who do not. Finally, initial firm productivity levels and amenities are largely dispersed in the cross-section of firms: the estimates of σ_z and σ_a are 2.13 and 0.90, respectively.

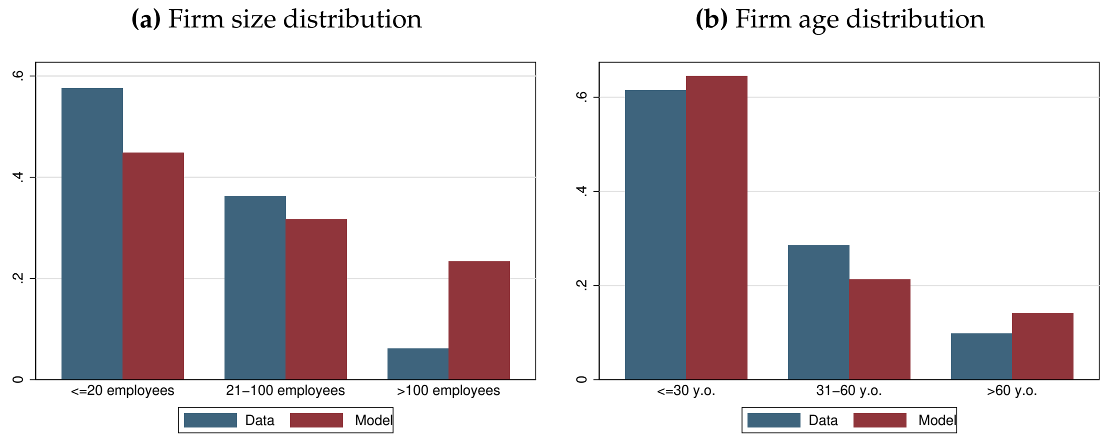
The standard errors on these estimates are small, suggesting the empirical moments are informative about the estimated parameters. To check that correct identification is achieved, Figure C.2 in Appendix C.2 shows that each targeted moment moves significantly and monotonically in response to changes in the parameter they are meant to inform.[16]

The model also replicates the empirical firm size and age distributions observed in the Netherlands despite neither being part of the targeted moments. Panel A of Figure 7 reports the percent of firms belonging to different firm size bins, in the data (blue bars) and the model (red bars). In the data, about 57.6 percent of the observed firms have less than 20 employees, while only around 6.2 percent of them employ more than 100 employees. The model reproduces this pattern. Panel B reports the percentage of firms across different firm age groups in the data and the model. In both cases, around 60 percent of firms are under 30 years old, while around 10 percent of them are over 60 years old.

[15]The estimation loss is minimized across all dimensions in the parametric space. See Figure C.1 in Appendix C.1.

[16]In Appendix C.2, we also show that i) the cross-contamination in the estimates is limited (Figures C.3, C.4, and C.5); and ii) the parameters estimates are not sensitive to the calibrated value of the revenue elasticity of labor, ξ (Figure C.6).

Figure 7: Non targeted moments - Model vs. Data



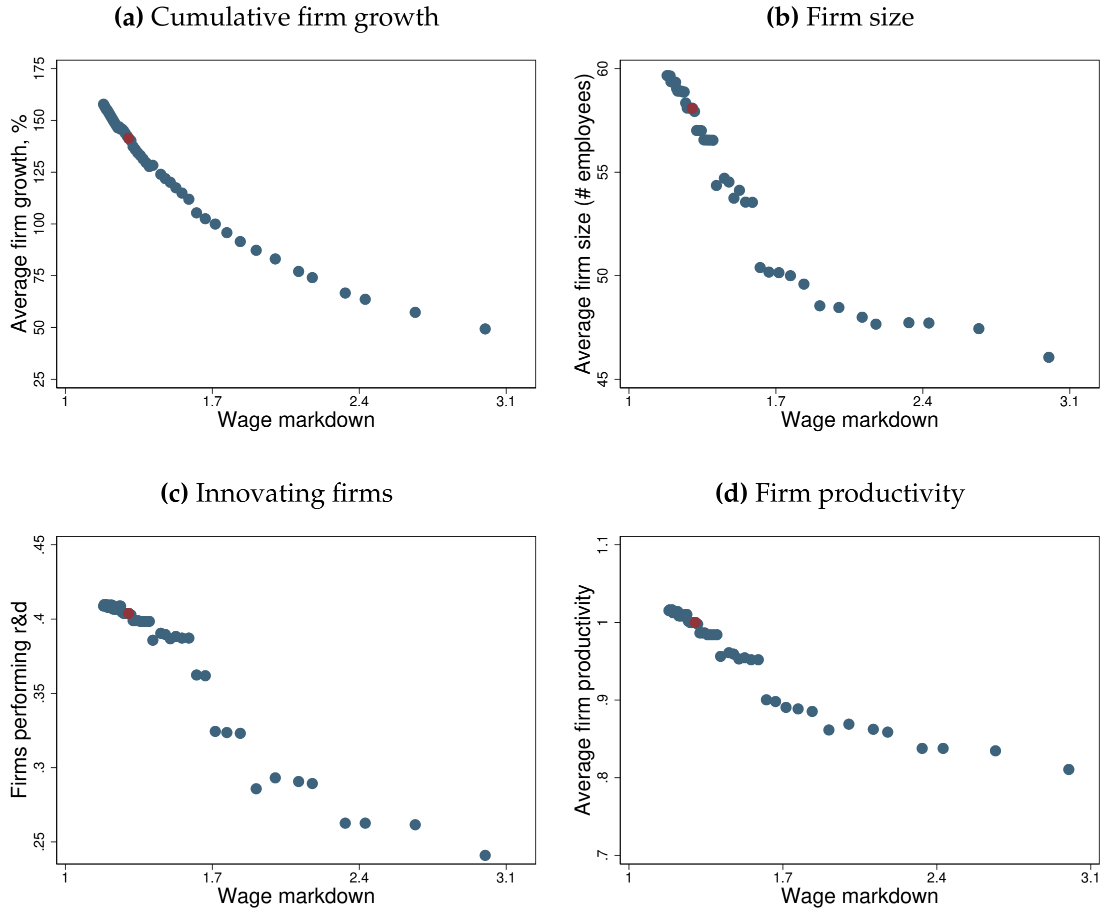
NOTES: Blue bars refer the shares of firms over firm size groups (panel A) and firm age groups (panel B) observed in the data. Red bars refer to the corresponding shares predicted by the model.

5 Labor market power and firm dynamics

We are now ready to discuss how labor market power affects firm dynamics and aggregate productivity. To this end, we construct counterfactual economies that differ from the benchmark only with respect to their labor supply elasticity while leaving all other parameters unchanged. As a result, the counterfactual economies are replicas of the Netherlands, except for differences in ϵ^L. In the benchmark economy, the labor supply elasticity is equal to 3.318, a value chosen to match a median markdown of 1.301. In the counterfactual economies, we let the elasticity vary between 0.6 and 5.5. These values correspond to wage markdowns ranging from 1.18 to 3, approximately the same values we estimate for our sample of middle- and high-income countries in Section 2.

Figure 8 reports the average cumulative life cycle firm growth (panel A), the average firm size (panel B), the share of firms adopting productivity-enhancing technologies (panel C), and the average firm productivity (panel D), for counterfactual economies with different degrees of labor market competition. The red dot refers to the benchmark economy, the Netherlands. The blue dots refer to counterfactual scenarios.

Figure 8: Firm Dynamics, Technology Adoption, and Labor Market Power



NOTES: The red circle refers to the Netherlands. Blue circles refer to counterfactual economies differing in their labor supply elasticity. Panel A reports the average cumulative firm size growth. Panel B reports the average firm size. Panel C reports the share of firms adopting productivity-enhancing technology. Panel D reports the average firm productivity, relative to the baseline (normalized to 1).

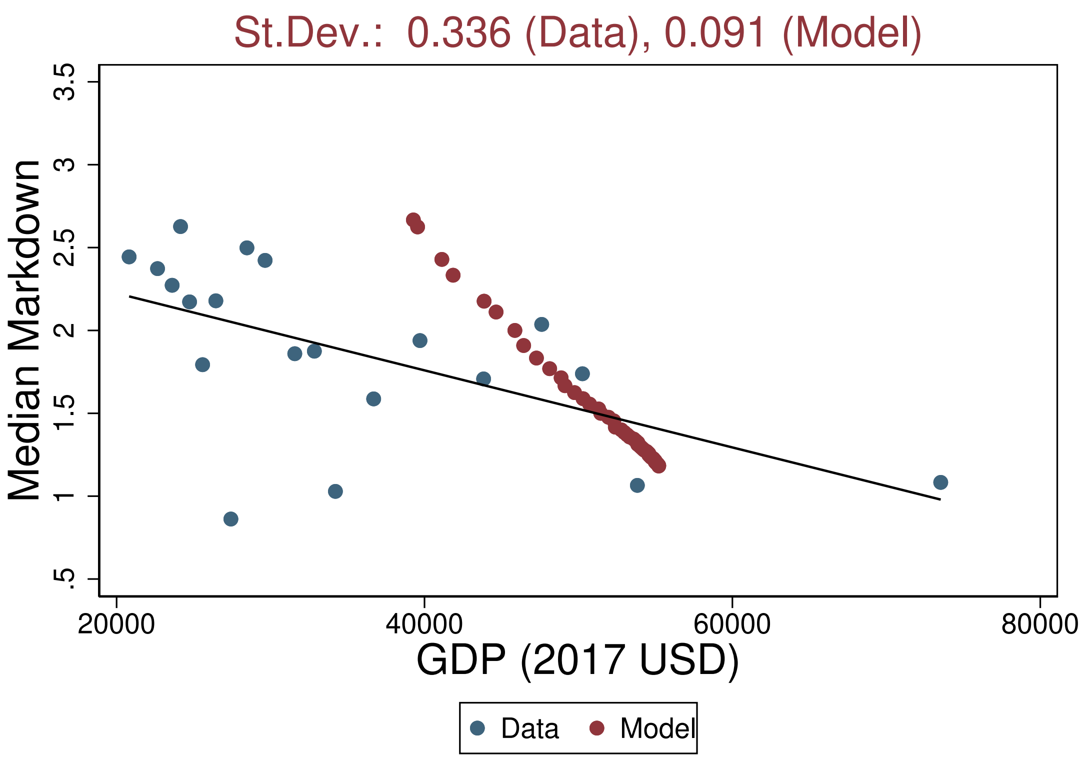
The model is consistent with the evidence described in Section 2. Labor market power affects firm dynamics over the life cycle. Reducing labor market competition leads to a significant reduction in the cumulative firm size growth. As wage markdown increases from 1.2 to 3, the average firm growth rate shrinks by more than a half, from 150 to about 50 percent. Lower firm growth reduces the average firm size (panel B). As labor supply elasticity reduces, the average firm size drops from approximately 60 to around 45 employees.

Like in the data, firms also are more likely to innovate when the labor market is more competitive (panel C). As we increase the labor supply elasticity, the share of firms investing in productivity doubles from 25 to 40 percent. As a result, the average firm productivity is higher (by about 20 percent) in more competitive labor markets (panel D).

5.1 Cross-country income differences

How important is labor market power in generating output dispersion in our sample of countries? We can answer this question through the lens of our model.

Figure 9: Cross-Country Income Differences: Model vs. Data



NOTES: Blue dots refer to the observed GDP p.c. from panel A of Figure 5. Red dots refer to the model-based predicted GDP p.c. obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both observed and predicted GDP p.c. are expressed in USD at 2017 constant price. SOURCE: WDI and authors' calculation.

Figure 9 scatters the observed GDP per capita of each country in our sample against the estimated wage markdown (blue dots) and compares that to simulated GDP per capita obtained in counterfactual economies that feature the labor supply elasticity in line with our estimates of wage markdown reported in Section 2 (red dots). As before, all other parameters are kept fixed at their benchmark values. Both observed and simulated GDP per capita are reported in USD at 2017 constant price level. Notice that, despite considering only middle and high-income countries, the observed differences in GDP per capita are significant, and amount up to a factor of 3.

A few comments are in order. First, there is a positive correlation between simulated and observed GDP per capita across countries. This is because a lower labor supply elasticity generates a high wage markdown, which slows down firm dynamics, reduces efficiency, and lowers aggregate output. On the other hand, the model generates less variation in output than is observed in the data: through the lens of our model, the observed cross-country variation in the estimated markdown only explains a portion of the GDP dispersion observed in our sample.

To quantify the contribution of labor market power, we compute the standard deviation in model-based and observed (log) GDP per capita. We find values of 0.09 and 0.34, respectively. We interpret it as the ability of the model to account for about 27 percent (=0.091/0.336*100) of the observed variation in GDP per capita across countries.[17] This value is similar to the estimates of GDP losses caused by size-dependent distortions (Restuccia and Rogerson, 2008; Bento and Restuccia, 2017; Tamkoç and Ventura, 2024), and is larger than gains from reducing firms' labor market power obtained using static models of imperfect competition (Berger et al., 2022; Amodio et al., 2025b; Armangué-Jubert et al., 2025).

[17] Similar results are obtained when we compare measured TFP across countries against model-based predicted GDP per capita. See Figure C.7 in Appendix C.3.

5.2 The role of productivity-amenity correlation.

Our quantitative results rely on the assumption that initial entrepreneurial productivity is independent of amenities. In Appendix C.5, we assess the robustness of our findings by relaxing this assumption. To do so, we solve a version of the model where new entrants draw their initial productivity and amenities from a bi-variate zero-mean log-normal distribution, $\Psi(z, a)$, with variances equal to σ_z^2 and σ_a^2, and correlation σ_{za}.

We estimate σ_{za} through indirect inference, i.e., we choose σ_{za} to match the correlation between firms' (log) wage and job amenity documented in Sockin (2024). They estimate firm-specific (log) wage and satisfaction premia using matched employer-employee data on wages and employer ratings in the United States, and find a positive (conditional) correlation of 0.622 (Table 2, column 3): satisfaction improves significantly at higher-paying firms. We extend our set of target to include this correlation, and estimate σ_{za} to be 0.296. We report further estimation details in Appendix C.5.

When initial entrepreneurial productivity and amenities are positively correlated, the counterfactual experiments generate a standard deviation in model-based (log) GDP per capita of 0.101. That is, the model is able to account for about 30 percent (=0.101/0.336*100) of the observed variation in GDP per capita across countries—see Figure C.10 in Appendix C.5. This value aligns to the one obtained in the main analysis (i.e., 27 percent), suggesting our findings extends to scenarios where σ_{za} is not zero.

5.3 The role of entry and investment costs.

In the baseline version of the model, operating and investment costs are both fixed output costs. Klenow and Li (2025) argue against this assumption. They document that average employment per firm increases with the level of overall labor productivity, both over time and across space, and argue that specifying costs in terms of final goods in models of firm dynamics might produce implications that are at odds with this evidence. By contrast, they suggest denominating costs in terms of labor.

We test the robustness of our results to expressing costs in terms of labor. To do so, we assume that c_f and c_x are both overhead labor costs entrepreneurs need to incur to operate in the industry and to innovate. In Appendix C.6, we report value functions, definition of equilibrium, and estimation details.

We then use this model to conduct the same analysis as in Section 5.1. That is, we generate a series of counterfactual economies that feature the labor supply elasticity in line with our estimates of wage markdown reported in Section 2, and compared the simulated GDP per capita of these economies to the observed GDP per capita of each countries in our sample—see Figure C.11 in Appendix C.6.

Expressing operating and innovation costs in terms of labor significantly amplifies the importance of labor market power. The standard deviation in model-based (log) GDP per capita is 0.17. That is, the model can account for about 50 percent (=0.174/0.336*100) of the observed variation in GDP per capita across countries, about twice as much of the variation explained by the baseline model.

This happens because, by increasing surviving firm productivity on average, fixed output costs shrink the mass of operating firms at the margin that can be moved by changing labor market power, thereby reducing the productivity gains of having a more competitive labor market. If costs are defined in terms of labor, the scope for reallocation is larger: the same increase in labor market competition can drive a larger measure of low-productivity, high-amenity out of market.

5.4 Robustness

Imputed firm-level moments. The WB-ES data only provides information for firms with at least 5 employees, making the sample biased towards relatively larger firms. In Appendix C.7, we re-estimate our baseline model using firm-level moments that are imputed to cover the entire span of *existing firms* in the Netherlands.

Our findings are robust. We find that the standard deviation in model-based (log) GDP per capita is 0.08— see Figure C.12 in Appendix C.5. This version of the model can explain about 24 percent (=0.082/0.336*100) of the observed variation in GDP per capita across countries, a value only slightly smaller than the one obtained in the main analysis.

Alternative identification. Finally, we test how robust our findings are to an alternative identification strategy and re-estimate the baseline model using a different set of moments. Specifically, we make the model over-identified by targeting the shares of firms with different size (≤ 20, 21-100, and > 100 employees), the shares of firms of different ages (≤ 30 y.o., between 30-60 y.o., and > 60 y.o.), and the shares of firms investing in R&D by number of employees. Moreover, we replace the average cumulative firm growth rate with the average annualized growth rate. Details of the estimation are reported in Appendix C.8.

Expanding the set of targets only slightly alter our counterfactual results. With a standard deviation in model-based (log) GDP per capita of 0.06, the over-identified model can still explain about 19 percent (=0.064/0.336*100) of the observed cross-country variation in GDP per capita—see Figure C.13 in Appendix C.8.

6 Mechanisms

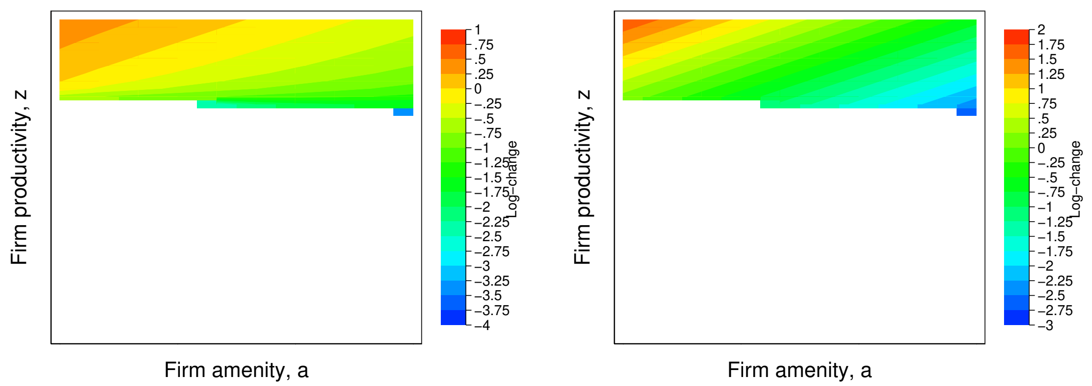
In this section, we shed light on the model mechanisms behind the outcomes presented in the previous sections. To keep the discussion compact, we compare the benchmark economy (Netherlands) with a single counterfactual economy, featuring the same degree of labor market power observed in Greece. This choice is motivated by two reasons, i.e. i) Greece has one of the lowest GDP per capita in the sample, approximately one-half of that of the Netherlands (29,000 vs. 54,000 USD); and ii) the degree of labor market competition is much weaker in Greece than the Netherlands: the estimated wage markdown is equal to 1.301 (vs. 2.623), corresponding to an elasticity of labor supply of 3.318 (vs. 0.616). To implement this experiment, we keep all parameters fixed at their benchmark values (Netherlands), and change the elasticity of labor supply to reproduce the median wage markdown estimated for Greece.

Compared to the Netherlands, the average size of firms is smaller in Greece (26 vs. 59 employees), firms grow less over the life cycle (84.5 vs. 153 percent), they are on average younger (the average age is 22.5 years vs. 30), and are less likely to invest in productivity innovation (18 vs. 41 percent). Differences in labor market competition can explain 29 percent of the differences in firm size, account for differences in average firm growth, and explain 27 and 74 percent of the differences in average firm age and share of firms investing in R&D, respectively (see Table C.3 in Appendix C.4).

Why does firm dynamics slow down when labor markets are less competitive? Figure 10 shows how firm-level employment (Panel A) and profits (Panel B) change when we move from a counterfactual economy with low labor market competition (Greece) to the benchmark economy with higher labor competition (Netherlands). Warmer-colored areas refer to firms with states (z_j, a_j) expanding in size and making higher profits when we increase the labor supply elasticity, ϵ^L. Cooler-colored areas refer to firms shrinking and losing profits.

Figure 10: Employment, Firms, and Investment Reallocation

(a) Profits **(b)** Employment



NOTES: Panel A shows the log-difference in firm-level profits for firms with different levels of firm productivity and amenities, between the benchmark economy, i.e. the Netherlands, and counterfactual economy, i.e. Greece. Panel B shows the log difference in firm-level employment between benchmark and counterfactual.

As discussed in Section 3.4, labor reallocates to high-productive, low-amenity firms and away high-amenities firms when labor market competition is stronger. As the elasticity of labor supply reduces, the relative importance of amenities increases, workers become less responsive to wage differences, and lack of competition pushes firms to post wages farther from the marginal revenue product of labor. As a result, profits increase for low-productivity high-amenity firms, allowing them to survive in the market, distorting allocative efficiency.

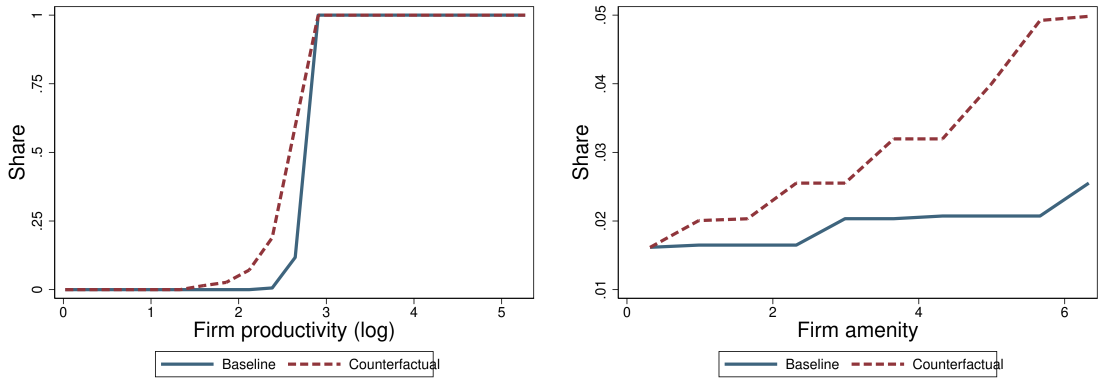
Changes in employment and profits also alter entrepreneurial and innovation decisions. Panels A and B of Figure 11 show the share of agents that choose to become entrepreneurs in the benchmark (blue line) and the counterfactual economy (red line), by levels of productivity and amenities. Similarly, Figure 12 compares the share of entrepreneurs that adopt productivity-enhancing technologies by levels of productivity (panel A) and amenities (panel B) in the benchmark (blue line) and the counterfactual economy (red line). In Appendix C.4 we report the associated policy functions for entrepreneurship and technology adoption.

In the model, agents choose whether to become entrepreneurs or wage workers aware of the equilibrium labor supply and demand curves. When ϵ^L is low, and the relative importance of amenities is higher, agents with low entrepreneurial productivity but high amenities can attract workers despite paying lower wages. Hence they anticipate being able to make net profits beyond what they could earn as wage workers, and choose to do so by self-selecting into entrepreneurship. Similarly, when ϵ^L is low and the role of productivity in shaping profits is weaker, returns to innovation diminish. This, together with the reallocation of profits towards high amenities and low-productivity firms, reduces dynamic investment, particularly for high-productivity firms. By favoring high-productivity firms, labor market competition operates a skill-biased force.

Figure 13 describes the overall distortionary effects of labor market power. It reports the simulated (log) average revenue product of labor (APL) for firms with different levels of productivity (panel A) and amenities (panel B) in the baseline (blue line) and counterfactual (red dashed line) economies. In a fully undistorted economy, the average revenue product of labor would not vary across firms. Instead, in both scenarios it increases with productivity and declines with amenities. At the same time, it changes much less steeply with productiv-

Figure 11: Entrepreneurs

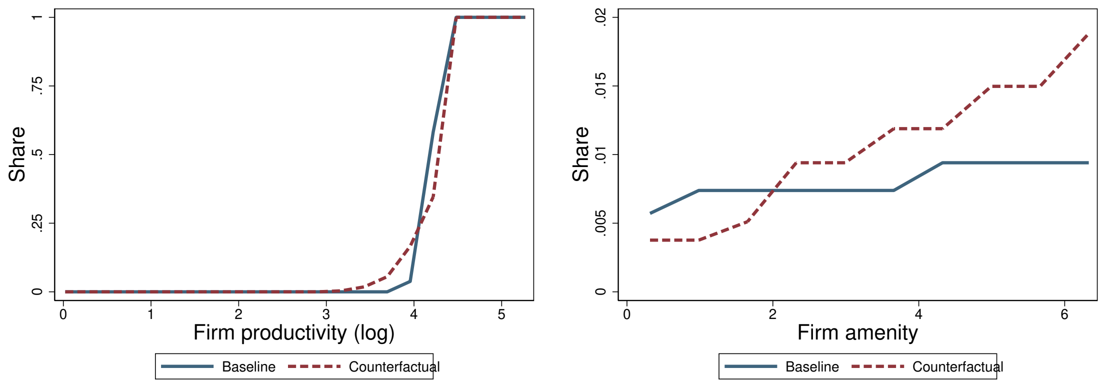
(a) By productivity **(b)** By amenities



NOTES: This figure shows the share of agents that become entrepreneurs by level of (log) productivity (Panel A) and level of amenities (Panel B) in the benchmark (blue line) and counterfactual (red dashed line) economy. The benchmark economy refers to the Netherlands. The counterfactual economy refers to an economy where ϵ^L is chosen to match the median markdown observed in Greece (leaving other parameters to their benchmark values).

Figure 12: Firms Adopting Productivity-Enhancing Technology

(a) By productivity **(b)** By amenities

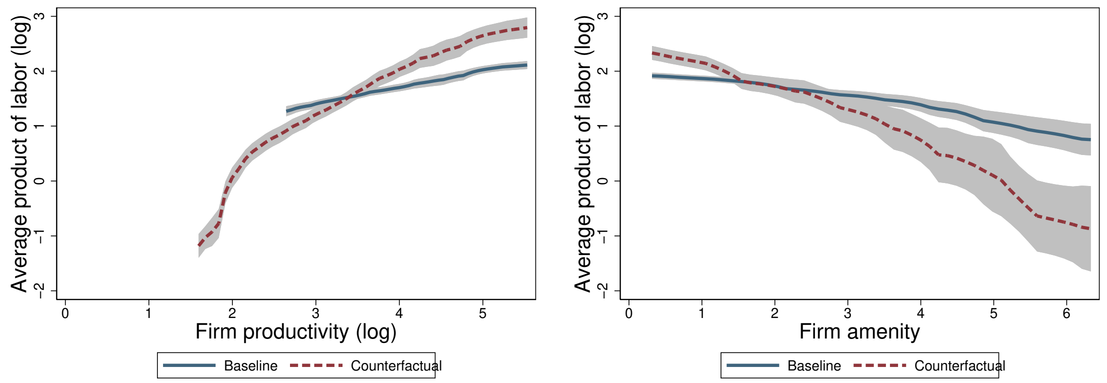


NOTES: This figure shows the share of entrepreneurs that adopt productivity-enhancing technologies by levels of (log) productivity (panel A) and amenities (panel B) in the benchmark (blue line) and counterfactual (red dashed line) economy. The benchmark economy refers to the Netherlands. The counterfactual economy refers to an economy where ϵ^L is chosen to match the median markdown observed in Greece (leaving other parameters to their benchmark values).

ity and amenity when labor market is more competitive: the elasticities of the APL with respect to productivity and amenities are 0.31 and -0.28 in the baseline economy, respectively, versus 0.79 and -0.81 in the counterfactual.

Figure 13: Revenue product versus productivity and amenities

(a) By productivity **(b)** By amenities



NOTES: This figure reports the simulated average revenue product of labor (in logs) for firms with different levels of productivity (panel A) and amenities (panel B) in the baseline economy (blue line) and counterfactual scenario (red dashed line). The shaded areas refer to 95 percent confidence interval constructed using simulated data.

Finally, we decompose how much each margin, meaning, allocation of labor, selection into entrepreneurship, and technology adoption, contribute to differences in output per capita between benchmark and counterfactual. We do it using two alternative experiments. In the first one, we change ϵ^L to the level observed in Greece while fixing entry and investment policy functions to the ones in the benchmark economy, i.e. we construct a counterfactual economy keeping selection into entrepreneurship and technology adoption fixed. In the second alternative scenario, we perform the same exercise while fixing only the entry policy function from the benchmark, i.e. we construct a counterfactual economy keeping only selection into entrepreneurship at the benchmark level. Table C.4 in Appendix C.4 reports the full outcomes of this decomposition.

Around 39 percent of the losses to income per capita induced by higher labor market power can be attributed to changes in employment allocation, keeping innovation policy and selection into entrepreneurship unchanged. About 56 percent of the output losses are explained by the distortion to innovation policy, while the remaining 5 percent can be attributed to distorted selection into entrepreneurship. These findings bridge the gap between the estimates of the cost of labor market power (Berger et al., 2022; Amodio et al., 2025b; Armangué-Jubert et al., 2025) and the dynamic inefficiency cost studied in models of misallocation (Restuccia and Rogerson, 2017; Guner et al., 2018; Guner and Ruggieri, 2022). By altering investment and firm growth, our results suggest that losses from labor market power may be greater than those estimated by previous studies that focus solely on the static labor allocation effect.

7 Conclusion

This paper studies how labor market power affects firm dynamics and aggregate efficiency across countries. We do it using a general equilibrium model of labor market monopsony featuring endogenous entrepreneurial choice, dynamic firms' technology adoption, and taste shocks for employers that endow them with wage-setting power. Calibrated to the Netherlands, the model reproduces cross-country differences in firm growth, technology adoption, and firm age structure through changes in labor supply elasticity.

Through the lens of the model, we find that the differences in labor market power can explain 27 percent of differences in income per capita across middle- and high-income countries, and no less than 19 percent over multiple robustness checks. Moreover, two-thirds of the explained dispersion can be attributed to distorted selection into entrepreneurship and lower innovation rates. Losses from labor market power may be greater than those estimated by previous studies that focus solely on static labor allocation effects.

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Supplementary Appendix

A Data Appendix

A.1 Sample summary and descriptive statistics

Table A.1 lists the countries in the sample, their survey waves, and the number of firm-level observations recorded.

Table A.1: Harmonized WBES sample merged with GDP p.c.

Country	Survey Waves	Num. Obs.
Austria	2021	600
Bahamas, The	2010	150
Belgium	2020	614
Croatia	2007 2013 2019 2023	1871
Cyprus	2019	240
Denmark	2020	995
Estonia	2009 2013 2019 2023	1257
Finland	2020	759
France	2021	1566
Germany	2021	1694
Greece	2018 2023	1198
Hungary	2009 2013 2019 2023	2237
Ireland	2020	606
Israel	2013	483
Italy	2019	760
Kazakhstan	2009 2013 2019	2590
Latvia	2009 2013 2019	966
Lithuania	2009 2013 2019	904
Luxembourg	2020	170
Malaysia	2015 2019	2221
Malta	2019	242
Netherlands	2020	808
Poland	2009 2013 2019	2366
Portugal	2019 2023	2069
Romania	2009 2013 2019 2023	2842
Russian Federation	2009 2012 2019	6547
Saudi Arabia	2022	1573
Slovak Republic	2009 2013 2019	972
Slovenia	2009 2013 2019	955
Spain	2021	1051
Sweden	2014 2020	1191

SOURCE: WBES, WDI, and authors' calculation.

Table A.2 reports summary statistics for the variables used in Section 2.

Table A.2: Summary statistics

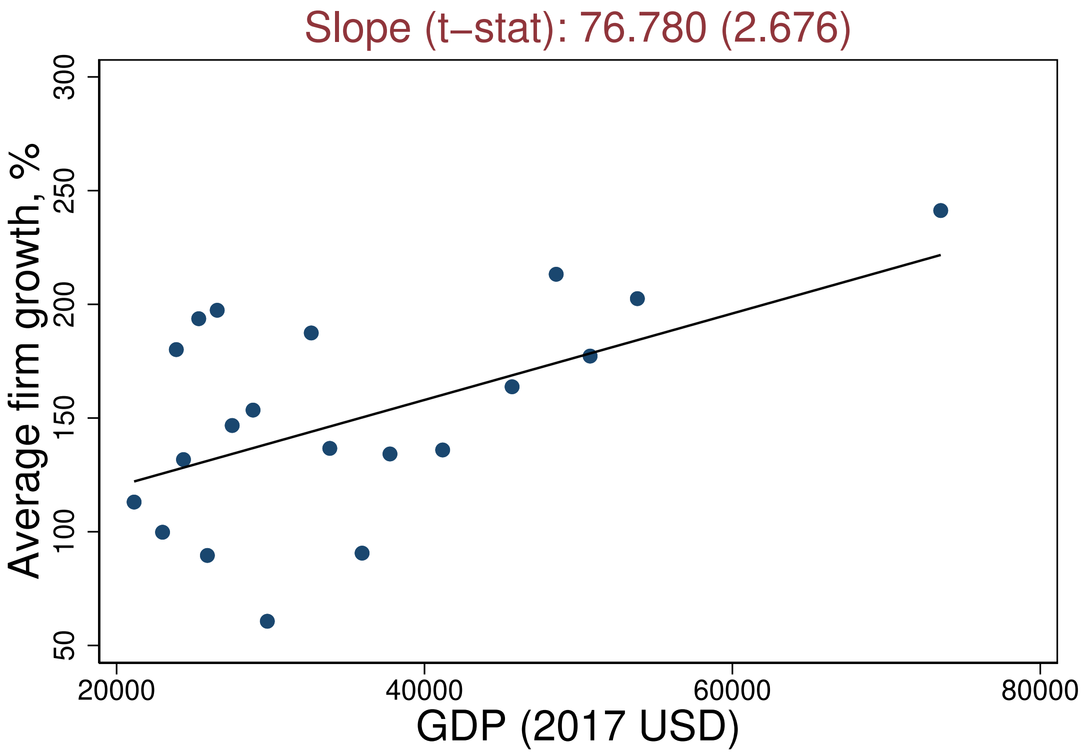
Statistics	Mean	Median	SD	P25	P75
Firm size	57.949	48.069	32.042	36.075	71.117
Firm size cumulative growth, %	113.50	114.07	31.51	91.284	135.04
Firm size annualized growth, %	7.5342	7.3405	2.2200	5.9854	8.7939
Firm age (years)	21.418	19.631	7.4695	15.901	28.770
Firm performing R&D	0.2230	0.1975	0.1461	0.0917	0.3130
Firm performing process innovation	0.3118	0.2830	0.1797	0.1755	0.4354
Firm performing product innovation	0.4183	0.4082	0.2022	0.2357	0.5657
Sales (log)	14.018	13.976	0.7352	13.572	14.570
Material expenditure (log)	12.726	12.684	0.9570	12.091	13.504
Capital expenditure (log)	12.665	12.614	0.8514	11.969	13.437
Foreign	0.1120	0.1203	0.0608	0.0610	0.1501
Shareholding	0.7213	0.8302	0.2873	0.5375	0.9511

NOTES: Firm size to the current number of employees. Firm size cumulative growth is computed as the log difference between the current number of employees and the number of employees recorded in the first year of operations. Firm size annualized growth is computed dividing firm size cumulative growth by the recorded firm age. Sales refer to the establishment's total annual revenues. Material expenditure refers to the cost of raw materials and intermediate goods used in production in the last fiscal year. Capital expenditure refers to the cost for the establishment to re-purchase all of its machinery. Sales, material, and capital expenditure are deflated using the US GDP deflator and expressed in 2009 USD. Foreign is a dummy variable taking value 1 if the firm is owned by private foreign individuals, companies or organizations by any percent, 0 otherwise. Shareholding is a dummy variable taking value 1 if the firm is shareholding company (including both those with traded and non-traded shares), 0 otherwise. SOURCE: WBES and authors' calculation.

A.2 Firm Growth across Countries

Figure A.1 complements Figure 1 from the main text, and it reports the average cumulative firm size growth for firms that are 40 years old across countries. The average firm growth is higher in richer countries, even when conditioned on firm age, and it increases from around 75 percent in countries with a GDP per capita of 20,000 USD to more than 150 percent in countries with a GDP per capita above 60,000 USD.

Figure A.1: Average cumulative firm size growth, 40 years

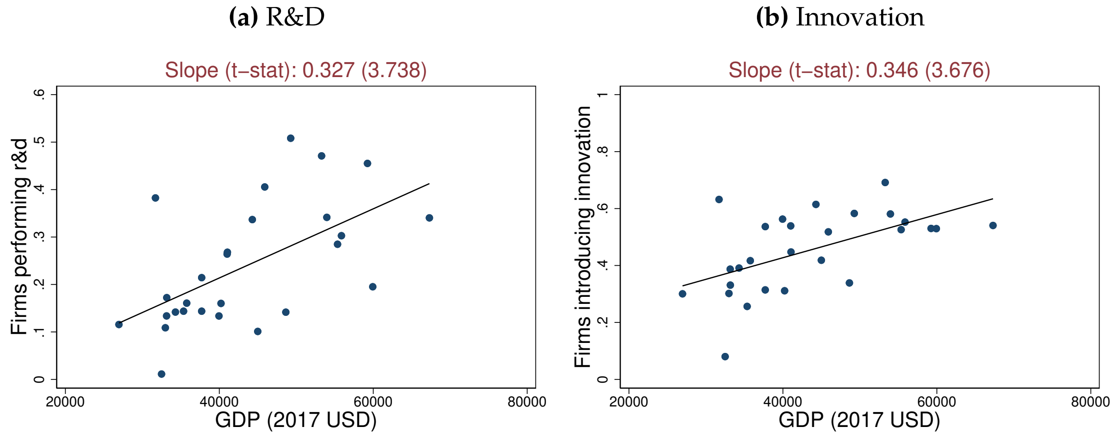


NOTES: This figure is a binscatter of the country-average cumulative firm size growth after 40 years of operations over GDP per capita. The fitted line is obtained from an auxiliary regression on GDP per capita. GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation.

A.3 Technology Adoption across Countries

Here we complement the evidence described in Section 2 on technology adoption across countries. To this purpose, we use data from the Community Innovation Survey (CIS). The CIS, is conducted in every European Union (EU) Member State to collect data on firm-level R&D and innovation in businesses, i.e. on product innovation (goods or services) and process innovation (e.g. production methods; logistics; information processing and communication; accounting and administrative operations; organizational, management and marketing aspects). We use country-wide aggregated statistics provided by the Eurostat for 2022.

Figure A.2: Firm technology adoption over development



NOTES: Panel A binscatters the share of firms that conduct R&D across countries over their GDP per capita. Panel B binscatters the share of firms that conduct innovation (both product and process) across countries over their GDP per capita. Fitted lines are obtained from auxiliary regressions on GDP per capita. GDP per capita is expressed in 2017 USD constant prices. SOURCE: CIS, EUROSTAT, WDI, and authors' calculation.

Figure A.2 binscatters the share of firms performing R&D (panel A) and the share of firms conducting innovation (panel B), including product and process innovations. This figure replicates the evidence on technology adoption discussed in Section 5.1 in the main text. As we move from low to high GDP per capita countries in the EU, the share of firms performing R&D increases from 0.10 to 0.50. Similar, the share of firms conducting either product or process innovation (or both) doubles from 0.30 to 0.6.

A.4 Wage markdown across Countries

We measure labor market power at the firm level by comparing the firm's marginal revenue product of labor to the wage paid (Amodio et al., 2024). To do so, we first assume the following revenue function specification,

$$\ln y_{it} = \alpha + \xi \ln \ell_{it} + \gamma \ln k_{it} + \delta \ln m_{it} + \omega_{it} + \epsilon_{it}$$

where y_{it} is firm sales, ℓ_{it} denotes number of employees, k_{it} is capital, m_{it} materials of firm i and time t. Finally, ω_{it} captures a combination of productivity differences across firms and demand-side factors affecting the output price, while ϵ_{it} is instead an unobserved i.i.d. idiosyncratic shock to revenues with mean zero.

We estimate the parameters of the revenue production function separately for each country in the sample using a control function approach as in Levinsohn and Petrin (2003). This method relies on three main assumptions: (i) the term ω_{it} evolves according to a first-order Markov process; (ii) the term ω_{it} is the only unobservable in the firm's input demand function; and (iii) the input demand function is invertible in ω_{it}. Under these three assumptions, we can control for unobserved productivity and demand shocks non-parametrically, using materials as proxy variables. Let

$$m_{it} = F_M(k_{it}, \omega_{it})$$

be the firm-level demand function for materials. As commonly assumed, capital k_{it} is an argument of this function because it is assumed to be already determined when materials are chosen. Assuming monotonicity of F_M, one can invert the materials demand function and express productivity ω_{it} as a function of capital, k_{it}, and material, m_{it}, i.e.,

$$\omega_{it} = \Lambda_M(k_{it}, m_{it}).$$

Substituting the latter back into the production function equation, we can write:

$$\ln y_{it} = \alpha + \xi \ln \ell_{it} + \phi(k_{it}, m_{it}) + \epsilon_{it}, \tag{13}$$

where

$$\phi(k_{it}, m_{it}) = \gamma \ln k_{it} + \delta \ln m_{it} + \Lambda_M(k_{it}, m_{it}).$$

As in Levinsohn and Petrin (2003), we can obtain an estimate of the revenue elasticity of labor, $\hat{\xi}$, by estimating equation (13) with OLS, using a fully interacted second-order polynomial in k_{it} and m_{it} to approximate $\phi(k_{it}, m_{it})$.

We estimate equation (13) separately for every country in the sample. Table A.3

reports selected summary statistics for the distribution of estimated country-specific revenue elasticities of employment, $\hat{\xi}$.

Table A.3: Estimated revenue elasticity of employment

Statistics	Mean	Median	SD	P25	P75
$\hat{\xi}$	0.4472	0.4110	0.2042	0.3267	0.5557

NOTES: This table reports summary statistics for the distribution of estimated country-specific revenue elasticities of employment, $\hat{\xi}$. SOURCE: WBES and authors' calculation.

The mean estimate for the elasticity is 0.4472. The distribution of estimates is not excessively dispersed across countries: the standard deviation is 0.2042, and the inter-quartile range is 0.2283.

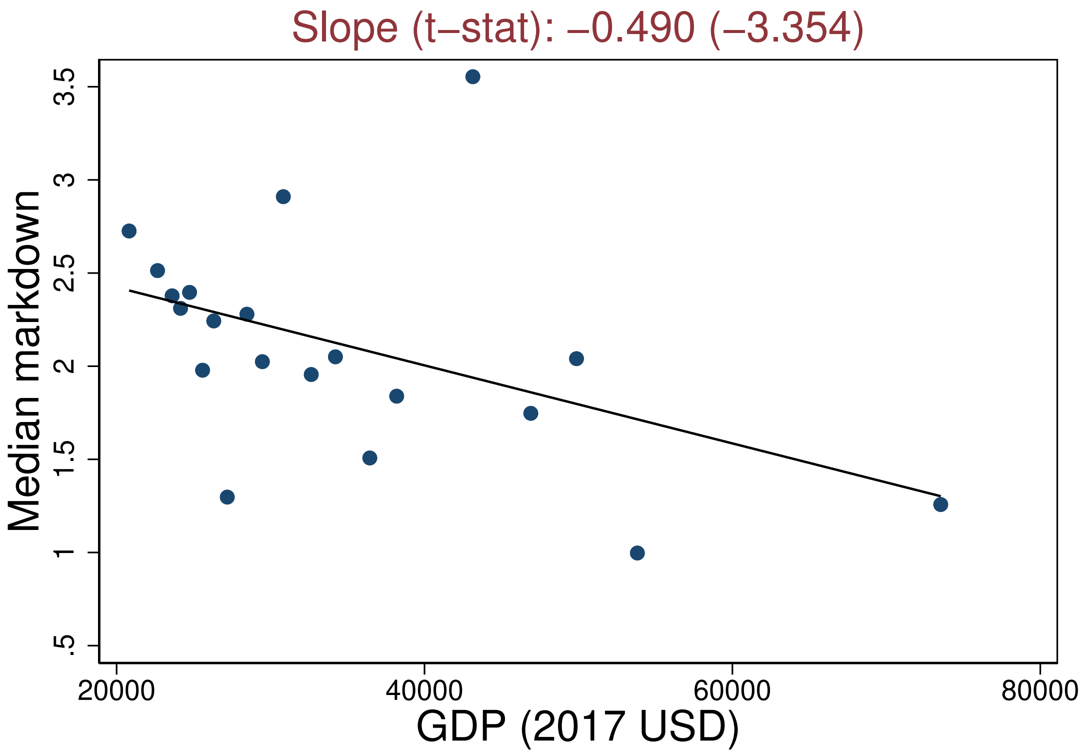
Using our estimates for the revenue elasticity of labor, $\hat{\xi}$, we derive the wage markdown as a ratio between the marginal revenue product of labor and the wage paid by firm i at time t,

$$\mu_{it} = \frac{\text{MRPL}_{it}}{w_{it}} \quad \text{where} \quad \text{MRPL}_{it} = \frac{\partial y_{it}}{\partial \ell_{it}} = \hat{\xi}\frac{y_{it}}{\ell_{it}},$$

Notice that if the product market were not perfectly competitive, firms could raise their output price above marginal costs. On the other hand, price markup will not confound the estimate for wage markdown obtained from the equation above because it is based on the revenue elasticity of labor (Amodio et al., 2024).

Finally, we complement the analysis of labor market power by estimating firm-level markdowns controlling for year fixed effects and firm-level observables, such as location fixed effects, 2-digit ISIC v.3.1 industry fixed effects, firm-age fixed effects, a dummy for foreign ownership, and a dummy for shareholding companies. Figure A.3 binscatters the estimated median markdown against GDP per capita of countries in the sample. Controlling for firm-level observables does not alter the cross-country patterns uncovered in Section 2.

Figure A.3: Median markdown across countries - Controlling for firm-level observables.



NOTES: This figure shows a binscatter of the median firm-level markdown across countries over GDP per capita. Firm-level markdown are estimated controlling for year fixed effects and firm-level observables, such as location fixed effects, 2-digit ISIC v.3.1 industry fixed effects, age fixed effects, a dummy for foreign ownership, and a dummy for shareholding companies. Fitted lines are obtained from an auxiliary regression on log GDP per capita. In the caption of each panel, we report the estimated slope of the regression and the t-stat (in parentheses). GDP per capita is expressed in 2017 USD constant prices. SOURCE: WBES, WDI, and authors' calculation. SOURCE: WBES, and authors' calculation.

B Model Appendix

B.1 Derivation of labor supply function

Equation (3) in the main text shows the that labor supply function to a firm j is equal to

$$L_j = L \int_{\mathcal{Z}\times\mathcal{A}} p_{ij}\phi(z_i, a_i) dz_i da_i$$

where p_{ij} is the probability that a worker i chooses to work for a firm j, while $\phi(z_i, a_i)$ is the equilibrium distribution of workers i across productivity and amenities.

Recall that p_{ij} is equal to:

$$p_{ij} = \frac{\exp\left(\frac{U(z_i,a_i,z_j,a_j)}{\sigma_\nu}\right)}{\int_L^1 \exp\left(\frac{U(z_i,a_i,z_k,a_k)}{\sigma_\nu}\right) dk}$$

By a change of variable and expanding the value functions, we can re-write the previous expression as:

$$p_{ij} = \frac{\exp\left(\frac{\epsilon^L \ln(w_j)+a_j+\beta(1-\delta_w)\mathbb{E}_{z_i'}\max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_\nu}\right)}{E\int_{\mathcal{Z}\times\mathcal{A}} \exp\left(\frac{\epsilon^L \ln(w_k)+a_k+\beta(1-\delta_w)\mathbb{E}_{z_i'}\max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_\nu}\right)\psi(z_k,a_k)dz_k da_k}$$

Substituting the last expression for p_{ij} into the labor supply function, L_j, we can write:

$$L_j = L\int_{\mathcal{Z}\times\mathcal{A}} \left(\frac{\exp\left(\frac{\epsilon^L \ln(w_j)+a_j+\beta(1-\delta_w)\mathbb{E}_{z_i'}\max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_\nu}\right)}{E\int_{\mathcal{Z}\times\mathcal{A}} \exp\left(\frac{\epsilon^L \ln(w_k)+a_k+\beta(1-\delta_w)\mathbb{E}_{z_i'}\max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_\nu}\right) d\psi(z_k,a_k)}\right) d\phi(z_i,a_i)$$

or,

$$L_j = L \int_{\mathcal{Z}\times\mathcal{A}} \left(\frac{\exp\left(\frac{\epsilon^L \ln(w_j)+a_j}{\sigma_V}\right) \exp\left(\frac{\beta(1-\delta_w)\mathbb{E}_{z_i'} \max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_V}\right)}{E\int_{\mathcal{Z}\times\mathcal{A}} \exp\left(\frac{\epsilon^L \ln(w_k)+a_k+\beta(1-\delta_w)\mathbb{E}_{z_i'} \max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_V}\right) d\psi(z_k,a_k)} \right) d\phi(z_i,a_i)$$

which is equivalent to:

$$L_j = L\Theta \exp\left(\epsilon^L \ln(w_j) + a_j\right)$$

where Θ is an aggregate market shifter, defined as:

$$\Theta = \frac{1}{\exp(\sigma_V)} \int_{\mathcal{Z}\times\mathcal{A}} \left(\frac{\exp\left(\frac{\beta(1-\delta_w)\mathbb{E}_{z_i'} \max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_V}\right)}{E\int_{\mathcal{Z}\times\mathcal{A}} \exp\left(\frac{\epsilon^L \ln(w_k)+a_k+\beta(1-\delta_w)\mathbb{E}_{z_i'} \max\{V(z_i',a_i),\tilde{U}(z_i',a_i)\}}{\sigma_V}\right) \psi(z_k,a_k)dz_k da_k} \right) \phi(z_i,a_i)dz_i da_i$$

B.2 Equilibrium distribution

Equations (14), (15), and (16) describe the lows of motion for the measure of employers across productivity z and amenities a.

$$\begin{aligned}
\Omega(z_i,a)' = {} & (1-\delta_w)p_n[(1-\rho^z(z_{i-},a))\rho^e(z_{i-},a) + (1-\rho^e(z_{i-},a))]\Omega(z_{i-},a) \\
& + (1-\delta_w)p_i\rho^z(z_{i-},a)\rho^e(z_{i-},a)\Omega(z_{i-},a) \\
& + (1-\delta_w)(1-p_n)[(1-\rho^z(z_{i+},a))\rho^e(z_{i+},a) + (1-\rho^e(z_{i+},a))]\Omega(z_{i+},a) \\
& + (1-\delta_w)(1-p_i)\rho^z(z_{i+},a)\rho^e(z_{i+},a)\Omega(z_{i+},a) \\
& + \delta_w\Psi_z(z_i)\Psi_a(a)
\end{aligned} \tag{14}$$

$$
\begin{aligned}
\Omega(\overline{z}, a)' = {} & (1-\delta_w) p_n [(1-\rho^z(\overline{z}_-, a))\rho^e(\overline{z}_-, a) + (1-\rho^e(\overline{z}_-, a))]\Omega(\overline{z}_-, a) \\
& + (1-\delta_w) p_i \rho^z(\overline{z}_-, a)\rho^e(\overline{z}_-, a)\Omega(\overline{z}_-, a) \\
& + (1-\delta_w) p_n [(1-\rho^z(\overline{z}, a))\rho^e(\overline{z}, a) + (1-\rho^e(\overline{z}, a))]\Omega(\overline{z}, a) \\
& + (1-\delta_w) p_i \rho^z(\overline{z}, a)\rho^e(\overline{z}, a)\Omega(\overline{z}, a) \\
& + \delta_w \Psi_z(\overline{z})\Psi_a(a)
\end{aligned} \tag{15}
$$

$$
\begin{aligned}
\Omega(\underline{z}, a)' = {} & (1-\delta_w)(1-p_n)[(1-\rho^z(\underline{z}, a))\rho^e(\underline{z}, a) + (1-\rho^e(\underline{z}, a))]\Omega(\underline{z}, a) \\
& + (1-\delta_w)(1-p_i)\rho^z(\underline{z}, a)\rho^e(\underline{z}, a)\Omega(\underline{z}, a) \\
& + (1-\delta_w)(1-p_n)[(1-\rho^z(\underline{z}_+, a))\rho^e(\underline{z}_+, a) + (1-\rho^e(\underline{z}_+, a))]\Omega(\underline{z}_+, a) \\
& + (1-\delta_w)(1-p_i)\rho^z(\underline{z}_+, a)\rho^e(\underline{z}_+, a)\Omega(\underline{z}_+, a) \\
& + \delta_w \Psi_z(\underline{z})\Psi_a(a)
\end{aligned} \tag{16}
$$

B.3 Numerical algorithm

The algorithm to solve for equilibrium goes as follows:

1. Guess a stationary distribution of agents over productivity and amenities $\Omega(z, a)^1$.
2. Given the current distribution $\Omega(z, a)^i$:
   (a) Guess the entrepreneurship policy function $\rho^{e,j}(z, a)$.
   (b) Using $\Omega(z, a)^i$ and $\rho^{e,j}(z, a)$, compute the distributions of workers and entrepreneurs over z and a: $\phi(z, a)$ and $\psi(z, a)$, and the measures of workers, L, and entrepreneurs, E.
   (c) Given $\phi(z, a)$, $\psi(z, a)$, L and E, solve the fixed point of the value functions to obtain U, $\tilde{U}$, V, W and π.
   (d) Using V, and $\tilde{U}$, update $\rho^{e,j+1}(z, a)$.
   (e) Check for convergence of the entrepreneurship policy function, if not equal, return to step (2.b) with the new one.

3. Use Equations (14) and (15) and (16) to get $\Omega(z,a)^{i+1}$, if not sufficiently close to $\Omega(z,a)^{i}$ return to step 2.

B.4 Derivation of model properties

An interior solution to the static problem of the firm presented in Section 3 satisfies the following first order condition:

$$\xi z_j L_j^{\xi-1} = w_j \left(1 + \frac{1}{\epsilon^L}\right). \tag{17}$$

That is, the wage paid by firm j, w_j, is equal to a markdown, $(1+\frac{1}{\epsilon^L})^{-1}$ over its marginal product of labor, $\xi z_j L_j^{\xi-1}$. Taking logs, we get:

$$(1-\xi)\ln L_j = \ln z_j + \ln \xi - \ln w_j - \ln\left(1 + \frac{1}{\epsilon^L}\right).$$

Using the labor supply function to substitute for (log) wages, we obtain:

$$(1-\xi)\ln L_j = \ln z_j + \ln \xi - \left(\frac{1}{\epsilon^L}\ln L_j - \frac{1}{\epsilon^L}\ln L - \frac{1}{\epsilon^L}\ln \Theta - \frac{1}{\epsilon^L}a_j\right) - \ln\left(1 + \frac{1}{\epsilon^L}\right).$$

Solving for L_j, we get:

$$\ln L_j = \frac{\epsilon^L}{1+(1-\xi)\epsilon^L}\ln z_j + \frac{1}{1+(1-\xi)\epsilon^L}a_j + C \tag{18}$$

where

$$C = \frac{1}{1+(1-\xi)\epsilon^L}\left[\epsilon^L \ln\left(\frac{\epsilon^L}{1+\epsilon^L}\right) + \epsilon^L \ln \xi + \ln L + \ln \Theta\right].$$

Notice that the log difference between the employment of firms with different productivity levels, $\underline{z}$ and $\bar{z}$, and same amenities, a, is equal to:

$$\ln L(\bar{z},a) - \ln L(\underline{z},a) = \frac{\epsilon^L}{1+(1-\xi)\epsilon^L}[\ln(\bar{z}) - \ln(\underline{z})]$$

Similarly, the log difference between the employment of firms with the same productivity, z, and different amenities, $\underline{a}$ and $\bar{a}$, we get:

$$\ln L(z,\bar{a}) - \ln L(z,\underline{a}) = \frac{1}{1+(1-\xi)\epsilon^L}[\bar{a}-\underline{a}]$$

Taking the exponential of both log-differences gives equations (7) and (8) in the main text. This proves proposition 1.

Consider now the average product of labor for a firm j, i.e.,

$$\text{APL}_j = \frac{z_j L_j^{\xi}}{L_j}.$$

The elasticity of APL_j with respect to productivity z_j is equal to:

$$\frac{\partial \ln \text{APL}_j}{\partial \ln z_j} = \frac{\partial(\ln z_j + (\xi-1)\ln L_j)}{\partial \ln z_j} = 1 + (\xi - 1)\frac{\partial \ln L_j}{\partial \ln z_j}$$

Using equation (18), we know that

$$\frac{\partial \ln L_j}{\partial \ln z_j} = \frac{\epsilon^L}{1+(1-\xi)\epsilon^L}$$

which implies that

$$\frac{\partial \ln \text{APL}_j}{\partial \ln z_j} = 1 - \frac{(1-\xi)\epsilon^L}{1+(1-\xi)\epsilon^L}$$

Similarly, the elasticity of APL_j with respect to exponential of amenities, $\exp(a_j)$, is equal to:

$$\frac{\partial \ln \text{APL}_j}{\partial a_j} = \frac{\partial(\ln z_j + (\xi-1)\ln L_j)}{\partial a_j} = +(\xi - 1)\frac{\partial \ln L_j}{\partial \ln a_j}.$$

Using equation (18), we know that

$$\frac{\partial \ln L_j}{\partial a_j} = \frac{1}{1 + (1 - \xi)\epsilon^L}$$

which implies that

$$\frac{\partial \ln \mathrm{APL}_j}{\partial a_j} = -\frac{(1 - \xi)}{1 + (1 - \xi)\epsilon^L}$$

This proves proposition 2 in the main text.

Finally, we can use the first order condition in equation (17) to express variable profits for firm j, π_j, as:

$$\pi_j = z_j L_j^{\xi} - w_j L_j = w_j L_j \left(\frac{1 + \epsilon^L}{\xi \epsilon^L} - 1 \right)$$

This implies that the elasticity of variable profits, π_j, with respect to firm productivity, z_j, is equal to:

$$\begin{aligned} \frac{\partial \ln \pi_j}{\partial \ln z_j} &= \frac{\partial \left[\ln w_j + \ln L_j + \ln \left(\frac{1+\epsilon^L}{\xi \epsilon^L} - 1 \right) \right]}{\partial \ln z_j} \\ &= \frac{\partial \ln w_j}{\partial \ln z_j} + \frac{\partial \ln L_j}{\partial \ln z_j} \\ &= \frac{\partial \ln w_j}{\partial \ln z_j} + \frac{\epsilon^L}{1 + (1 - \xi)\epsilon^L} \end{aligned}$$

where the last line is obtained from differentiating equation (18) with respect to $\ln z_j$.

To obtain $\partial \ln w_j / \partial \ln z_j$, we need to solve for wages, $\ln w_j$. To do, we can plug $\ln L_j$ from equation (18) into the labor supply function in equation (3) of the main text, and obtain:

$$\begin{aligned}
\epsilon^L \ln w_j &= \ln L_j - a_j - \ln L - \ln \Theta \\
\epsilon^L \ln w_j &= \left(\frac{\epsilon^L}{1+(1-\xi)\epsilon^L} \ln z_j + \frac{1}{1+(1-\xi)\epsilon^L} a_j + C \right) - a_j - \ln L - \ln \Theta \\
\epsilon^L \ln w_j &= \frac{\epsilon^L}{1+(1-\xi)\epsilon^L} \ln z_j - \frac{(1-\xi)\epsilon^L}{1+(1-\xi)\epsilon^L} a_j + C - \ln L - \ln \Theta \\
\epsilon^L \ln w_j &= \frac{\epsilon^L}{1+(1-\xi)\epsilon^L} \ln z_j - \frac{(1-\xi)\epsilon^L}{1+(1-\xi)\epsilon^L} a_j + C - \ln L - \ln \Theta
\end{aligned}$$

which implies:

$$\begin{aligned}
\ln w_j = & \frac{1}{1+(1-\xi)\epsilon^L} \ln z_j - \frac{(1-\xi)}{1+(1-\xi)\epsilon^L} a_j \\
& + \frac{1}{1+(1-\xi)\epsilon^L} \left[\ln \xi + \ln \frac{\epsilon^L}{1+\epsilon^L} - (1-\xi) \ln L - (1-\xi) \ln \Theta \right].
\end{aligned}$$

Differentiating $\ln w_j$ with respect to $\ln z_j$, we get:

$$\frac{\partial \ln w_j}{\partial \ln z_j} = \frac{1}{1+(1-\xi)\epsilon^L}$$

Therefore:

$$\frac{\partial \ln \pi_j}{\partial \ln z_j} = \frac{1}{1+(1-\xi)\epsilon^L} + \frac{\epsilon^L}{1+(1-\xi)\epsilon^L} = \frac{1+\epsilon^L}{1+(1-\xi)\epsilon^L}$$

which is equal to equation (11) of the main text. Similarly, the elasticity of variable profits, π_j with respect to the firm amenity, $\exp(a_j)$, is equal to:

$$\begin{aligned}
\frac{\partial \ln \pi_j}{\partial a_j} = & \frac{\partial \left[\ln w_j + \ln L_j + \ln \left(\frac{1+\epsilon^L}{\xi \epsilon^L} - 1 \right) \right]}{\partial a_j} = \frac{\partial \ln w_j}{\partial a_j} + \frac{\partial \ln L_j}{\partial a_j} \\
= & -\frac{(1-\xi)}{1+(1-\xi)\epsilon^L} + \frac{1}{1+(1-\xi)\epsilon^L} = \frac{\xi}{1+(1-\xi)\epsilon^L}
\end{aligned}$$

which is equal to equation (12) of the main text. This proves proposition 3.

C Quantitative Appendix

C.1 Estimation

Table C.1 reports the list of parameters that are calibrated externally to the model, their values their values and targets/sources.

Table C.1: Parameters Set Without Solving the Model

Parameters	Description	Value	Targets/Source
A	Aggregate productivity shifter	1	normalization
σ_ν	Type-I GEV shock scale	1	normalization
β	Discount factor	0.961	annual interest rate=0.04
δ_w	Retirement rate	0.025	40 years in the labor market
ξ	Revenue elasticity of labor	0.333	production fct. estimation
ϵ^L	Elasticity of labor supply	3.318	median markdown=1.301

NOTES: The table shows the parameters calibrated externally, their values, and the target or source used.

Table C.2 reports the list of targeted moments and their moment counterparts in the benchmark estimation. The sum of squared deviations between empirical and simulated moments is equal to 1.7 percent.

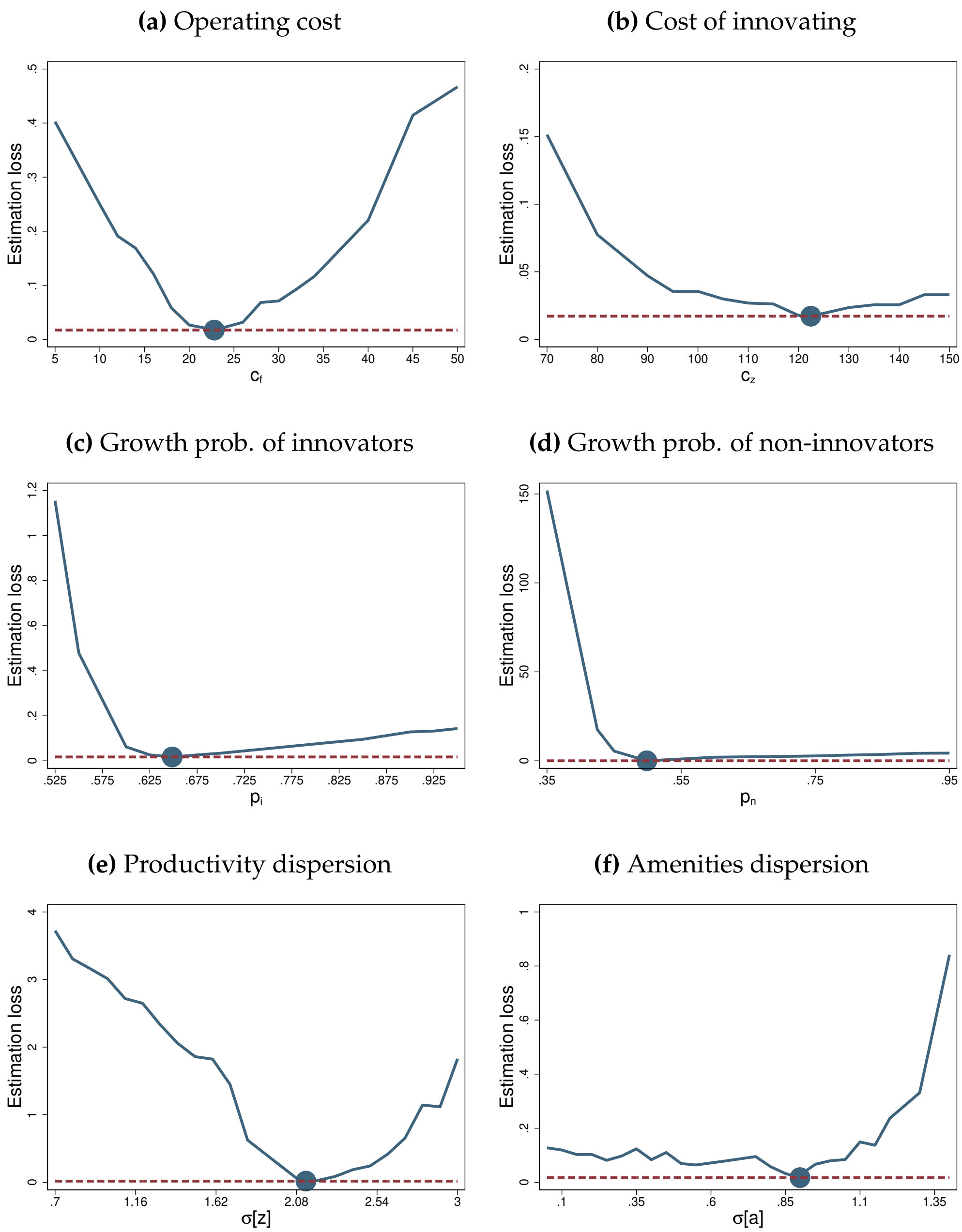
Table C.2: Targeted Moments: Data vs. Model

	Data (1)	Model (2)
Average firm size	59.02	58.09
Log firm size dispersion	0.988	1.091
Average cumulative growth rate (%)	152.6	141.4
Average firm age	30.13	30.84
Log wage dispersion	0.418	0.421
Firms investing in R&D	0.407	0.404

NOTES: This table reports the estimation fit. Column (1) the value of empirical targets. Column (2) reports model-based simulated moments.

Figure C.1 reports how the estimation loss varies over the parametric space, separately for each parameters. The blue dots refer to the point estimates. The red dashed line refers to the estimation loss evaluated at the final estimates. The estimation loss is minimized over each dimension of the parametric space.

Figure C.1: Estimation Loss



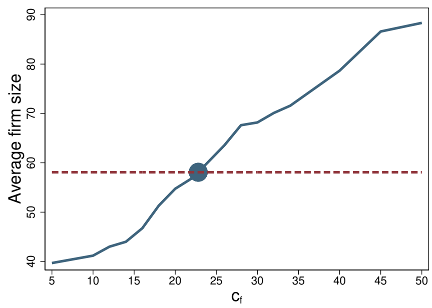
NOTES: This figure reports how the estimation loss (blue line) varies over the parametric space, separately for each parameters. The blue dots refer to the point estimates. The red dashed line refers to the estimation loss evaluated at the final estimates.

C.2 Identification

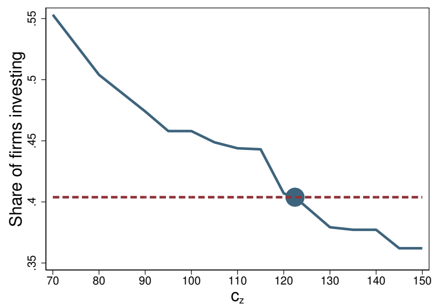
Figure C.2 reports how targeted moments change over the parametric space. Each panel shows that changing parameter values trigger a monotonic shift in the their respective target, suggesting identification is achieved.

Figure C.2: Mapping moments to parameters

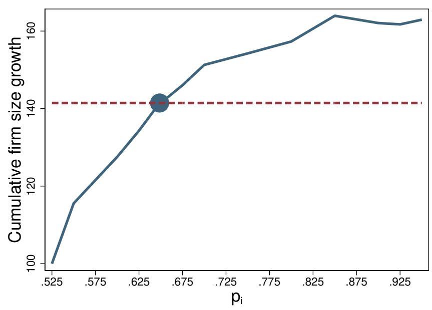
(a) Operating fixed cost



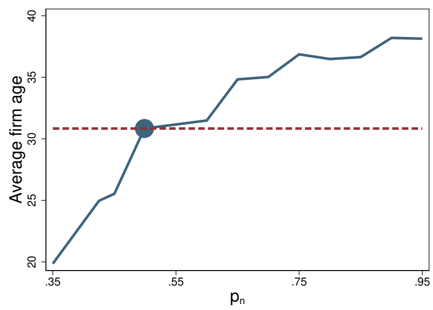
(b) Fixed cost of innovation



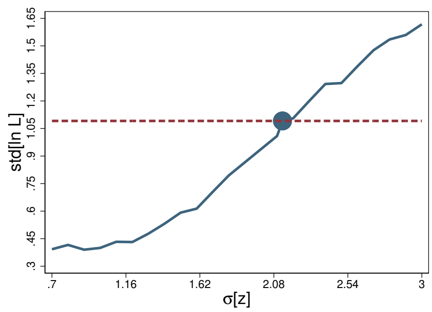
(c) Growth prob. of innovators



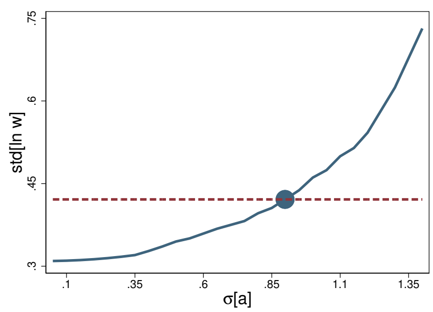
(d) Growth prob. of non-innovators



(e) Productivity dispersion



(f) Amenities dispersion



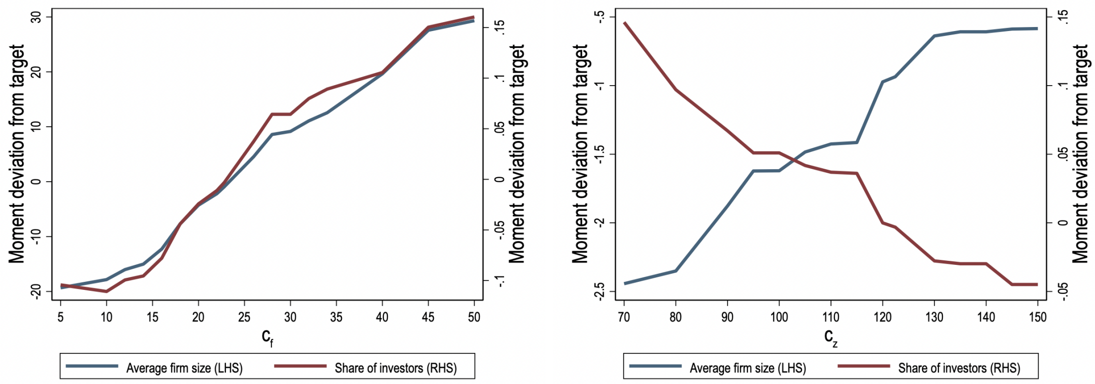
NOTES: This figure reports how each moment react to changes in parameters they are meant to identify. The red dashed line refer to the values of the empirical target.

Operating vs. innovation cost. In the estimation of the baseline model, the operating costs c_f is identified by targeting the average firm size (Figure C.2, panel A): by reducing profits, larger operating costs discourages lower-productivity, lower-amenities agents to become entrepreneurs. The innovation cost c_x is instead identified by targeting the share of firms conducting productivity-enhancing innovation (Figure C.2, panel B): everything else equal, larger fixed costs of innovation reduce the net benefits of innovation

Figure C.3 reports how both moments, i.e., the average firm size and the share of firms that innovate, respond to changes in c_f (panel A) and c_x (panel B), relative to the respective empirical targets.

Figure C.3: Contamination in the estimates of operating and innovation costs.

(a) Operating cost **(b)** Innovation cost



NOTES: This figure plots how the average firm size (blue line, left-hand side) and share of firms that innovate (red line, right-hand side) change with different value of c_f (panel A) and c_z (panel B), relative to their respective empirical targets.

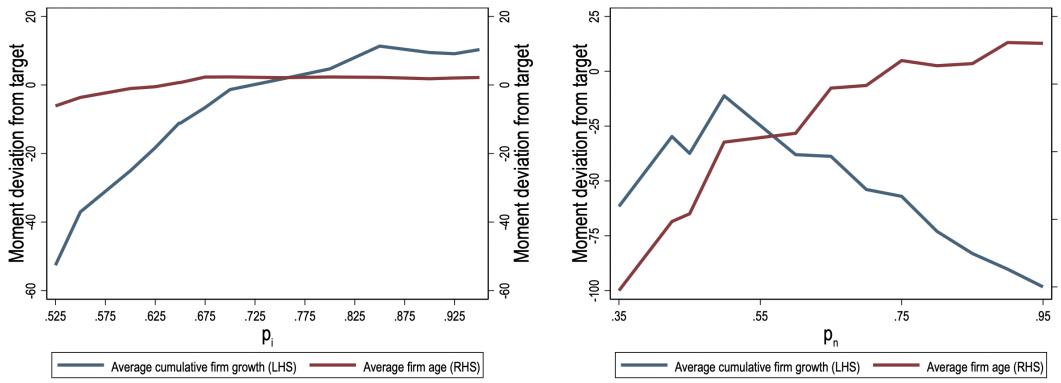
Changes in operating costs produce a monotonic increase in both moments. As we increase c_f, low-productivity, low-amenities agents are less likely to become entrepreneurs, and the distribution of active employer shifts towards entrepreneurs that can provide better amenities or with higher productivity. These entrepreneurs are also those who to run larger firms and more likely to engage in innovation. On the other hand, changes in innovation costs have an asymmetric effect: as we increase c_x, the share of firms that innovate declines, while the average firm size increases. While the first effect is trivial, the second one is again due to selection: when the innovation cost increases, average profits decline, and only high-productivity, high-amenities agents are have incentives to become entrepreneurs.

Productivity jumps: p_i vs. p_n. We repeat a similar analysis for the probabilities of productivity jump, p_i and p_n. In the estimation of the baseline model, these two parameters are identified by the average cumulative firm growth and the average firm age, respectively (Figure C.2, Panels E and F).

Figure C.4 reports how both moments, average cumulative firm size growth and average firm age, respond to changes in p_i (panel A) and $_n$ (panel B), relative to the respective empirical targets.

Figure C.4: Contamination in the estimates of productivity jumps

(a) Growth prob. of innovators **(b)** Growth prob. of non-innovators



NOTES: This figure plots how the average cumulative firm size growth (blue line, left-hand side) and the average firm age (red line, right-hand side) change with different value of p_i (panel A) and p_n (panel B), relative to their respective empirical targets.

Cross-contamination in the identification of p_i and p_n appears to be limited as well. Panel A shows that shifting p_i significantly affects the average cumulative firm size growth, i.e., the blue line rises monotonically, while it has a small effect on average firm age, i.e., the red line remains almost flat. This confirms that the former is informative about the parameter p_i, while the latter is not.

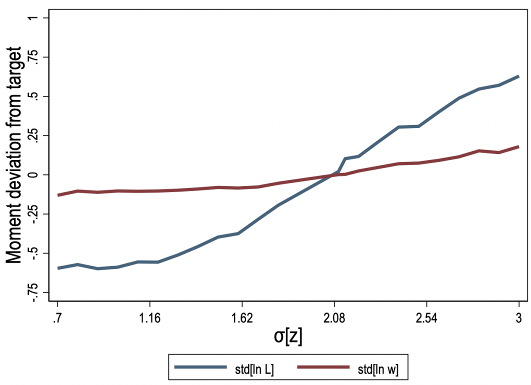
Viceversa, panel B shows that shifting p_n significantly affects the average firm age, i.e., the red line increases steeply (the right-hand axis), whereas changes in the average cumulative firm growth are large but not monotonic over the parametric space, i.e., the blue line first rises and then declines as we increase p_n. This confirms that the former moment is more informative about the parameters p_n compared to the latter.

Productivity dispersion vs. amenity dispersion. Finally, we study the identification of productivity and amenity dispersion, σ_z and σ_a. In the baseline estimation, σ_z is identified by targeting the standard deviation of log firm size, whereas σ_a is identified by the standard deviation of log wages (Figure C.2, Panels C and D). Higher probabilities rise the likelihood of productivity improvements over the firm life-cycle, increasing firm survival and employment growth.

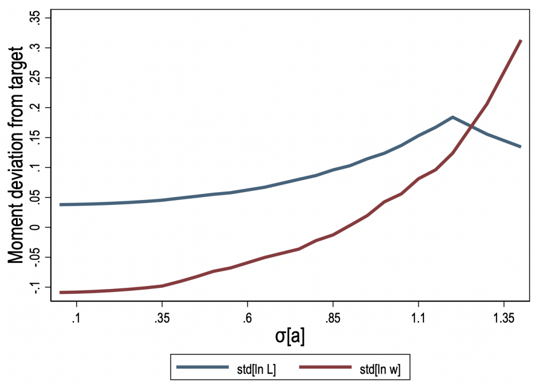
Figure C.5 reports how both moments, i.e., log size and log wage standard deviation, respond to changes in σ_z (panel A) and σ_a (panel B), relative to the respective empirical targets.

Figure C.5: Contamination in the estimates of productivity and amenity dispersion.

(a) Productivity



(b) Amenity



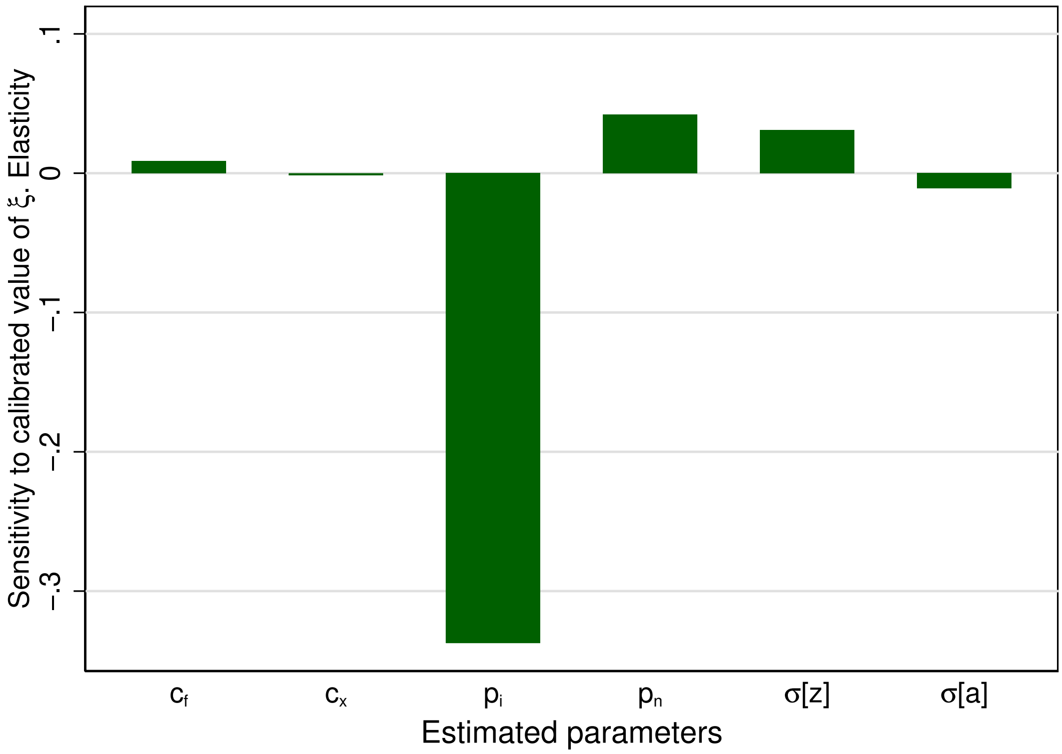
NOTES: This figure plots how the standard deviation of log firm size (blue line) and log wages (red line) change with different value of σ_z (panel A) and σ_a (panel B), relative to their respective empirical targets.

Cross-contamination in the identification of σ_z and σ_a appears to be limited. Panel A shows that shifting σ_z significantly affects firm size dispersion, i.e., the blue line rises steeply, while it only has a limited effect on wage dispersion, i.e., the red line remains almost flat. This confirms that the former is informative about the dispersion of productivity, while the latter is not.

Viceversa, panel B shows that shifting σ_a significantly affects wage dispersion, i.e., the red line increases steeply, whereas changes in firm size dispersion are not as much pronounced and are not monotonic over the parametric space, i.e., the blue line first rises by a little amount and then declines as we increase σ_a. This confirms that the former is more informative about the dispersion of amenities compared to the latter.

Sensitivity to the revenue elasticity. In the baseline model, we calibrate the revenue elasticity of labor ξ externally, using the corresponding estimate obtained in Section 2, i.e., 0.333. Here we assess how sensitive are the estimates to this assumption.

Figure C.6: Sensitivity to ξ



NOTES: This figure reports the sensitivity of each parameter estimates as in Jørgensen (2023) to the calibrated value of the revenue elasticity of labor, ξ.

We follow Jørgensen (2023) and compute a vector of sensitivity to ξ, defined as

$$S = -(J'WJ')^{-1}J'WD,$$

where W is the weighing matrix used in the estimation, J is the Jacobian matrix, whose $(i,j)^{\text{th}}$-entry is equal to $\partial d(\vartheta(i))/\partial\vartheta(j)$, and D is a vector of partial derivatives of moment conditions with respect to ξ, whose i^{th}-entry is $\partial d(\vartheta(i))/\partial\xi$.

Figure C.6 reports the elasticities of each estimated parameters k with respect to ξ; that is,

$$s(j) = S(j)\frac{\xi}{\vartheta(k)},$$

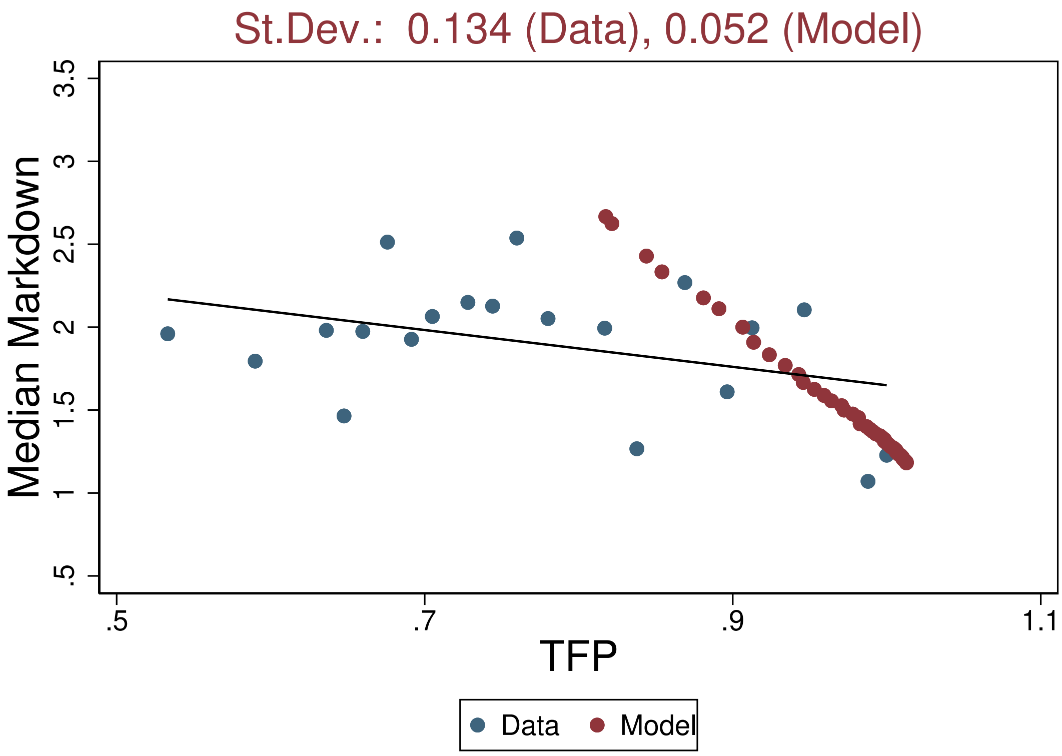
where $\vartheta(j)$ is the j^{th}-entry in the vector of estimated parameters, while $S(j)$ denotes the j^{th}-entry in the vector S.

The productivity jump for firms that innovate, p_i, is the most sensitive parameter: increasing the calibrated value of ξ by 100 percent would reduce the estimate of p_i by about 30 percent. This is because productivity-enhancing investments deliver a higher return to entrepreneurs when the revenue elasticity of labor is higher. As ξ increases, we require a lower productivity jump for firms that innovate, p_i, to match the same average cumulative firm size growth. We find that all the other parameters are relatively insensitive to the revenue elasticity of labor.

C.3 Cross-country TFP variation

Figure C.7 scatters the measured total factor productivity (TFP) of each country in our sample against the estimated wage markdown (blue dots) and it compares them to the simulated TFP obtained in counterfactual economies that feature the labor supply elasticity in line with our estimates of wage markdown reported in Section 2 (red dots). As before, all other parameters are kept fixed at their benchmark values. Both measured and simulated TFP are reported as fraction of the value measured for the Netherlands.

Figure C.7: Cross-Country TFP Differences: Model vs. Data



NOTES: Blue dots refer to measured TFP (variable *ctfp* in the Penn Word Table, v. 10.01). Red dots refer to model-based predicted GDP per capita, obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both variables are reported as fraction of the value measured in the Netherlands. SOURCE: PWT, WDI and authors' calculation.

The standard deviation in measured TFP across countries is equal to 0.134. This implies that the model accounts for about 39 percent (=0.052/0.134*100) of the observed variation in TFP across countries.

C.4 Counterfactual experiment: The Netherlands vs. Greece

Table C.3 reports various outcomes for benchmark economy (The Netherlands, in column 1) and counterfactual economy (Greece, in column 2), and compares the latter to their empirical counterparts (column 3).

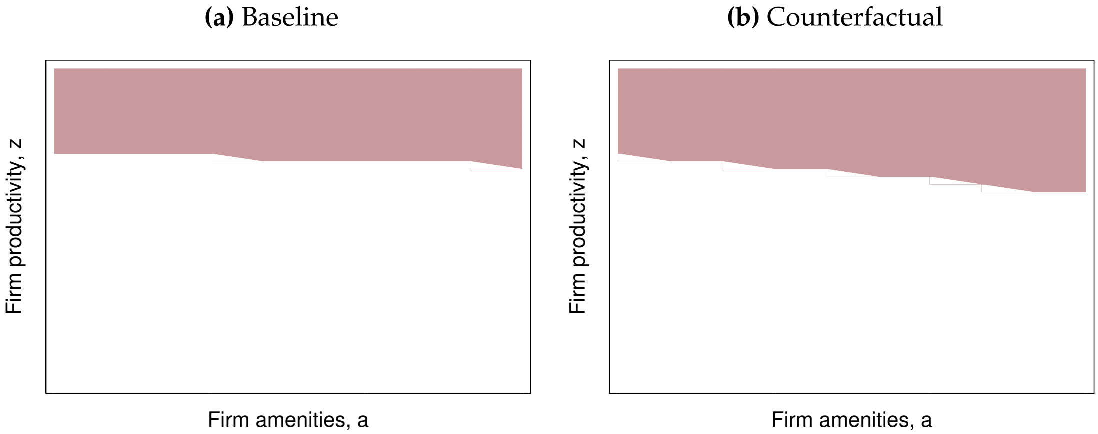
Table C.3: The Netherlands vs. Greece

	Netherlands Benchmark (1)	Greece Counterfactual (2)	Greece Data (3)	Explained (4)
Elasticity of labor supply	3.318	0.616	-	-
Average firm size	57.13	47.45	25.79	29.1%
Average cumulative growth rate (%)	142.3	58.4	84.5	123.3%
Average firm age	31.30	29.24	22.50	27.0%
Firms investing in R&D	0.432	0.262	0.177	73.9%
GDP p.c.	1	0.732	0.537	57.9%

NOTES: Column (1) reports the average life cycle firm growth, the average firm age, the share of entrepreneurs who innovate, and the value of GDP p.c. in the benchmark economy (Netherlands). Column (2) shows the same model-based moments in a counterfactual economy where ϵ^L is chosen to match the median markdown observed in Greece (leaving other parameters unchanged). Column (3) reports the empirical counterparts of these moments for Greece. Column (4) reports how much (in percent) of the difference between the Netherlands and Greece is explained by differences in labor supply elasticity, and it is computed as the differences between columns (1) and (2) in this table, divided by the difference between column (1) in table C.2 and column (3) in this table. GDP p.c. in column (4) is computed as the differences between columns (1) and (2) in this table, divided by the difference between columns (1) and (3) in this table.

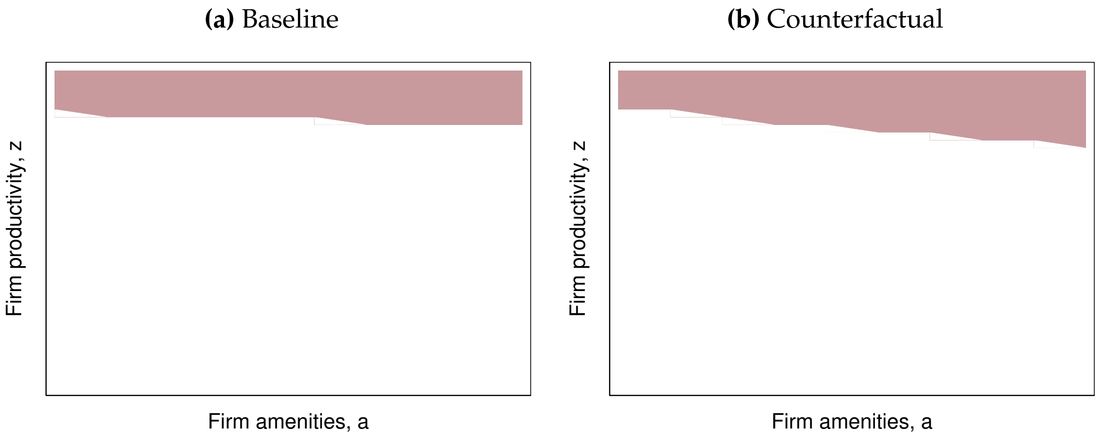
Figures C.8 and C.9 shows the policy functions for entrepreneurship and technology adoption in the baseline and counterfactual scenarios, i.e. The Netherlands vs. Greece. The colored areas refer to pairs of productivity and amenities where policy function takes value 1, while the white areas refers to to pairs of productivity and amenities where the policy functions takes value 0.

Figure C.8: Entrepreneurship Policy Functions



NOTES: Panels A and B show the policy function for entrepreneurship in the baseline and the counterfactual respectively. The colored areas refer to pairs of productivity and amenities where policy function takes value 1, while the white areas refers to to pairs of productivity and amenities where the policy functions takes value 0.

Figure C.9: Innovation Policy Functions



NOTES: Panels A and B show the policy function for investment in the baseline and counterfactual respectively. The colored areas refer to pairs of productivity and amenities where policy function takes value 1, while the white areas refers to to pairs of productivity and amenities where the policy functions takes value 0.

Table C.4 reports GDP per capita in the baseline scenario, i.e. the Netherlands (column 1); in a counterfactual scenario where the elasticity of labor supply is chosen to match the median markdown observed in Greece (column 4); in a scenario where the elasticity of labor supply is chosen to match the markdown observed in the baseline (the Netherlands), while keeping fixed the policy functions for investment at its counterfactual level (Greece), and leaving other parameters unchanged (column 3); and in a scenario where the elasticity of labor supply is chosen to match the markdown observed in the baseline (the Netherlands), while keeping fixed the policy functions for investment and entrepreneurship at their counterfactual levels (Greece), leaving other parameters unchanged (column 4).

Table C.4: Sources of output losses

	Baseline (1)	Baseline with counterfactual investment (2)	Baseline with counterfactual investment and entry (3)	Counterfactual (4)
ϵ^L	3.318	3.318	3.318	0.616
GDP p.c.	1	0.849	0.835	0.732
%	0	56.3	61.6	100

NOTES: Column (1) reports the value of GDP p.c. in the benchmark economy (the Netherlands). Column (4) shows the value of GDP p.c. in the counterfactual where we set ϵ^L to match the median markdown observed in Greece. Column (2) shows the value of GDP p.c. in the baseline economy (the Netherlands) when the investment policy is fixed at its counterfactual level (Greece). Column (3) shows the value of GDP p.c. in the baseline economy (the Netherlands) when both investment and entrepreneurship policy functions are fixed at their counterfactual level (Greece).

C.5 Role of productivity-amenity correlation.

In the baseline model, new entrants draw a tuple of characteristics (z, a) from two independent log-normal distributions, $\Psi_z(z)$, and $\Psi_a(a)$, with mean 0 and variances σ_z^2 and σ_a^2, respectively. In this section, we solve a version of the model where new entrants draw their initial productivity and amenities from a bivariate zero-mean log-normal distribution, $\Psi(z, a)$, with variances equal to σ_z^2 and σ_a^2, and correlation σ_{za}.

We estimate σ_{za} using indirect inference. Specifically, using the equilibrium solution of the model, we simulate a large panel of agents, and construct a model-based conditional correlation between firm-specific (log) wage and satisfaction premia, $\hat{\kappa}$, as

$$\hat{\kappa} = \frac{\text{cov}(\log w_{jt}, a_{jt})}{\text{var}(\log w_{jt})}$$

where a_{jt} and w_{jt} denote amenities and wages of firm j at time t in the simulated panel. Therefore, we choose σ_{za} such that $\hat{\kappa} = 0.622$. All other moments are kept the same as in the baseline version. Table C.5 reports the set of estimates, empirical moments used as targets and simulated counterparts. We obtain a value for σ_{za} of 0.296.

Endowed with new estimates, we repeat the same analysis of Section 5.1. Figure C.12 scatters the observed GDP per capita of each country in our sample against the estimated wage markdown (blue dots) and compares that to simulated GDP per capita obtained in counterfactual economies that feature counterfactual labor supply elasticities. As before, all other parameters are kept fixed at their benchmark values. Both measured and simulated GDP per capita are reported as fraction of the value measured for the Netherlands.

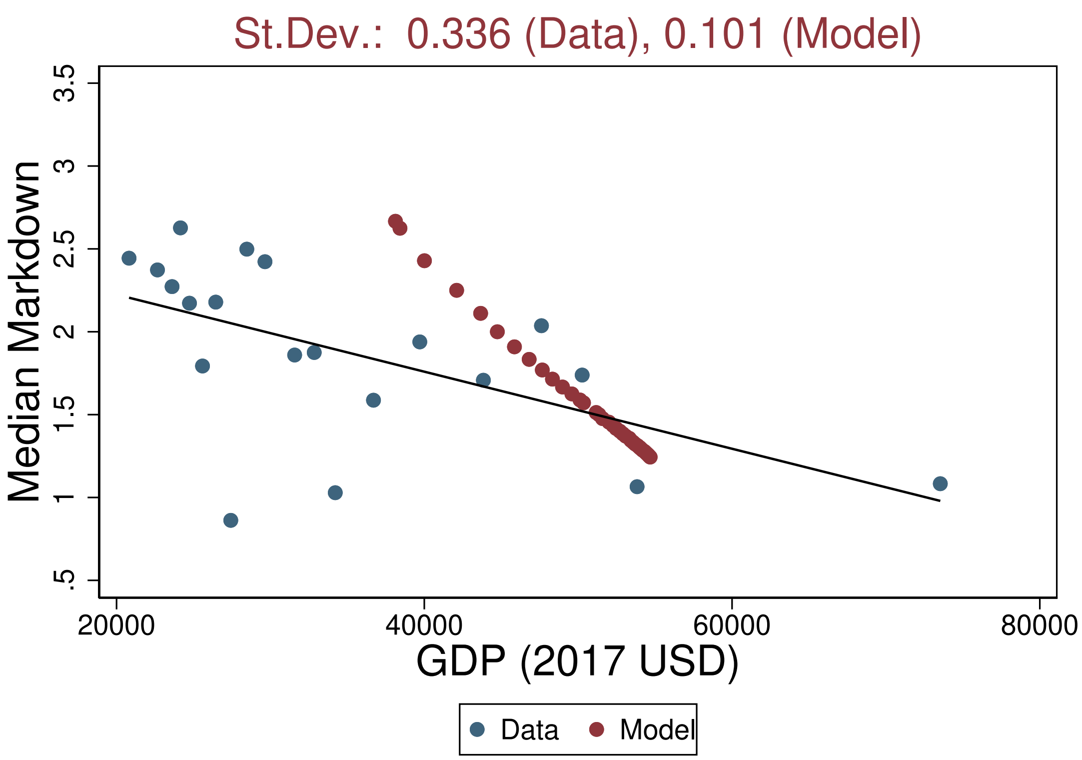
Using this version of the model, we find a standard deviation in model-based (log) GDP per capita of 0.10. Which implies that the model can explain about 30 percent (=0.101/0.336*100) of the observed variation in GDP per capita across countries.

Table C.5: Estimates and Targets - Model with Correlated Amenities

Parameters	Description	Estimates	Targets	Data	Model
c_f	Operating costs	78.97	Average firm size	59.02	57.95
c_x	Innovation costs	227.3	Share of firms investing in R&D	0.407	0.424
p_i	Growth prob. of innovators	0.588	Average cumulative growth rate (%)	152.6	140.9
p_n	Growth prob. of non-innovators	0.540	Average firm age	30.13	30.67
σ_z	Productivity dispersion	2.832	Log firm size dispersion	0.988	1.103
σ_a	Amenities dispersion	0.956	Log wage dispersion	0.418	0.425
σ_{za}	Correlation productivity-amenities	0.296	(log) Wage-amenities correlation	0.622	0.622

NOTES: The table shows the list of estimated parameters, their point estimates, the value of empirical targets, and model-based simulated moments. In this version of the model, the correlation between productivity and amenities is internally estimated to match the relation between firms' wage and job amenities estimated in Sockin (2024).

Figure C.10: Cross-Country Income Differences: Model with Correlated Amenities vs. Data



NOTES: Blue dots refer to the observed GDP p.c. from panel A of Figure 5. Red dots refer to the model-based predicted GDP p.c. obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both observed and predicted GDP p.c. are expressed in USD at 2017 constant price. In this version of the model, the correlation between productivity and amenities is internally estimated to match the relation between firms' (log) wage and job amenity estimated in Sockin (2024). SOURCE: WDI and authors' calculation.

C.6 Role of entry and investment costs.

In this section, we discuss a version of the model where both entry and investment costs are defined in terms of labor. In this model, the value to an agent i of being an entrepreneur is still equal to:

$$V(z_i, a_i) = \max\{V^I(z_i, a_i), V^N(z_i, a_i)\}$$

where $V^I(z_i, a_i)$ is the value of investing in productivity, equal to

$$\begin{aligned}
&V^I(z_i, a_i) = \epsilon^L \ln(\pi^I(z_i, a_i)) + a_i \\
&+ \beta(1-\delta_w)\big(p_i \max\{V(z_{i+}, a_i), \tilde{U}(z_{i+}, a_i)\} + (1-p_i)\max\{V(z_{i-}, a_i), \tilde{U}(z_{i-}, a_i)\}\big)
\end{aligned}$$

while $V^N(z_i, a_i)$ is value of not investing,

$$\begin{aligned}
&V^N(z_i, a_i) = \epsilon^L \ln(\pi^N(z_i, a_i)) + a_i \\
&+ \beta(1-\delta_w)\big(p_n \max\{V(z_{i+}, a_i), \tilde{U}(z_{i+}, a_i)\} + (1-p_n)\max\{V(z_{i-}, a_i), \tilde{U}(z_{i-}, a_i)\}\big).
\end{aligned}$$

On the other hand, as opposed to the baseline version, the profit flows net of fixed costs for entrepreneurs investing in productivity are the solution to the following problem:

$$\begin{aligned}
\max_{w_j} \pi_j^I(z_j, a_j) &= z_j L_j^{\xi} - w_j(L_j + c_z + c_f) \\
\textit{subject to} \quad L_j + c_z + c_f &= L\Theta \exp\left(\epsilon^L \ln(w_j) + a_j\right),
\end{aligned}$$

while the profit flows net of fixed costs for entrepreneurs who do not invest solve a different problem:

$$\begin{aligned}
\max_{w_j} \pi_j^N(z_j, a_j) &= z_j L_j^{\xi} - w_j(L_j + c_f) \\
\textit{subject to} \quad L_j + c_f &= L\Theta \exp\left(\epsilon^L \ln(w_j) + a_j\right).
\end{aligned}$$

All the other details of the model remain the same as in Section 3.

A stationary recursive equilibrium for this economy is a list of value functions $V(z_i, a_i)$, $U(z_i, a_i, z_j, a_j)$ and $\tilde{U}(z_i, a_i)$, an associated entrepreneurship policy function $\rho^e(z_i, a_i)$ and innovation policy function $\rho^z(z_i, a_i)$, a wage schedule $W(z_i, a_i)$, an allocation of labor $L(z_i, a_i)$, an aggregate measure of workers L, a distribution of agents over productivity and amenities, $\Omega(z_i, a_i)$, and distributions of wage workers and entrepreneurs over productivity and amenities, $\phi(z_i, a_i)$ and $\psi(z_i, a_i)$, such that:

- The labor supply to firm i, $L(z_i, a_i)$, satisfies equation (3);
- $\rho^e(z_i, a_i)$ and $\rho^z(z_i, a_i)$ solve the entrepreneurial and the innovation choices, and the value functions $V(z_i, a_i)$, $U(z_i, a_i, z_j, a_j)$ and $\tilde{U}(z_i, a_i)$ attain their maxima;
- The aggregate measure of workers is consistent with the entrepreneurial choices:

$$\begin{aligned} L &= \int_{\mathcal{Z}\times\mathcal{A}} [L(z_i, a_i) + c_f + c_z \rho^z(z_i, a_i)] \psi(z_i, a_i) dz_i da_i \\ &= \int_{\mathcal{Z}\times\mathcal{A}} (1 - \rho^e(z_i, a_i)) \Omega(z_i, a_i) dz_i da_i. \end{aligned}$$

- The distribution of agents over productivity and amenities, $\Omega(z_i, a_i)$ is stationary and replicates itself through entry and exit, and the policy functions, as in equations (14), (15) and (16), defined in Appendix B.2.
- The distributions of wage workers and entrepreneurs over productivity and amenities are stationary and defined as

$$\phi(z_i, a_i) = \frac{(1 - \rho^e(z_i, a_i))\Omega(z_i, a_i)}{\int_{\mathcal{Z}\times\mathcal{A}} (1 - \rho^e(z_i, a_i))\Omega(z_i, a_i) dz_i da_i},$$

  and

$$\psi(z_i, a_i) = \frac{\rho^e(z_i, a_i)\Omega(z_i, a_i)}{\int_{\mathcal{Z}\times\mathcal{A}} \rho^e(z_i, a_i)\Omega(z_i, a_i) dz_i da_i},$$

  respectively.

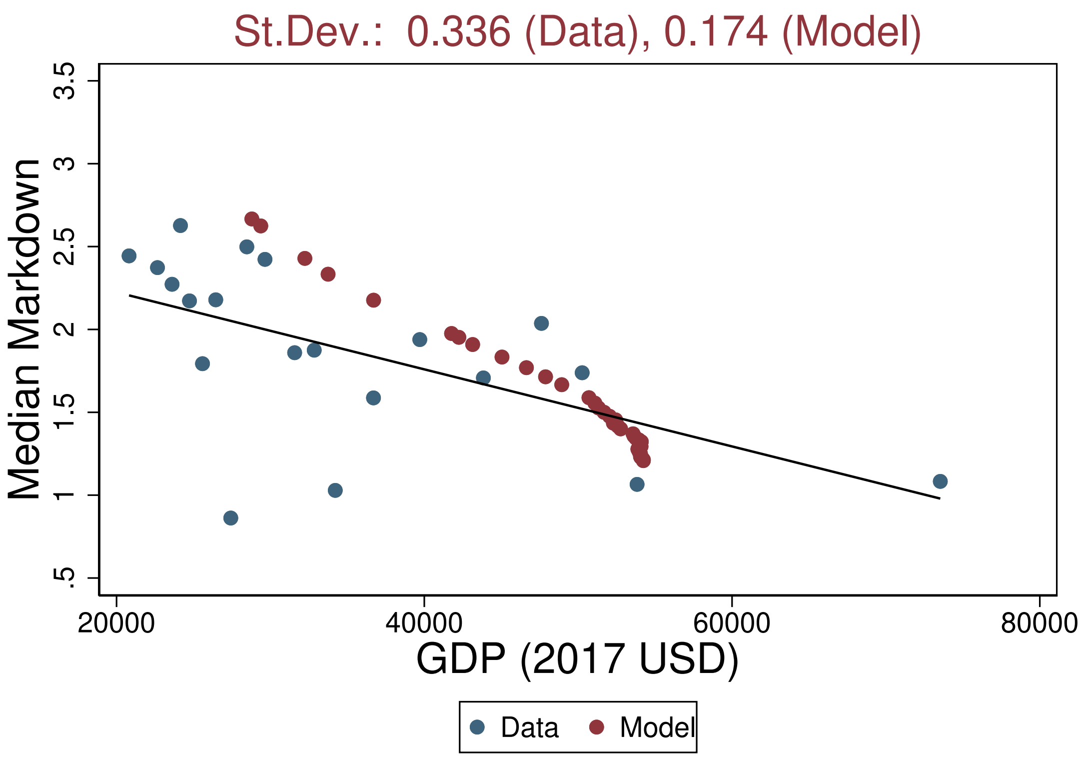
Table C.6 reports the set of estimates for this version of the model, empirical moments used as targets and simulated counterparts. Figure C.11 scatters the

Table C.6: Estimates and Targets - Model with Costs defined in terms of Labor

Parameters	Description	Estimates	Targets	Data	Model
c_f	Operating costs	2.274	Average firm size	59.02	58.10
c_x	Innovation costs	18.42	Share of firms investing in R&D	0.407	0.432
p_i	Growth prob. of innovators	0.603	Average cumulative growth rate (%)	152.6	150.9
p_n	Growth prob. of non-innovators	0.526	Average firm age	30.13	29.18
σ_z	Productivity dispersion	2.789	Log firm size dispersion	0.988	1.274
σ_a	Amenities dispersion	0.669	Log wage dispersion	0.418	0.475

NOTES: The table shows the list of estimated parameters, their point estimates, the value of empirical targets, and model-based simulated moments. In this version of the model, operating costs and investment costs are expressed in terms of labor.

Figure C.11: Cross-Country Income Differences: Model with Costs defined in terms of Labor vs. Data



NOTES: Blue dots refer to the observed GDP p.c. from panel A of Figure 5. Red dots refer to the model-based predicted GDP p.c. obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both observed and predicted GDP p.c. are expressed in USD at 2017 constant price. In this version of the model, operating costs and investment costs are expressed in terms of labor. SOURCE: WDI and authors' calculation.

observed GDP per capita of each country in our sample against the estimated wage markdown (blue dots) and compares that to simulated GDP per capita obtained in counterfactual economies that feature counterfactual labor supply elasticities. As before, all other parameters are kept fixed at their benchmark values. Both measured and simulated GDP per capita are reported as fraction of the value measured for the Netherlands.

Using this version of the model, we find a standard deviation in model-based (log) GDP per capita of 0.17. Which implies that the model can explain about 50 percent (=0.174/0.336*100) of the observed variation in GDP per capita across countries.

C.7 Estimation using imputed moments

The WB-ES data only provides information for firms with at least 5 employees, making the sample biased towards relatively larger firms. In this section, we re-estimate our baseline model using firm-level moments that are imputed to cover the entire span of *existing firms* in the Netherlands.

To this purpose, we impute the average firm size, the (log) firm size dispersion, and the share of firms investing in R&D as follows. The average firm size is imputed by fitting a Pareto mean to the sample average among *observed firms* from the WB-ES, imposing a left truncation at 5 employees. Specifically, let the firm size be a random variable following a Pareto distribution, with scale parameter $x_0 > 0$ and shape parameter $\alpha > 1$, i.e., firm size $\sim \text{Par}(x_0, \alpha)$. Then, its expected value is equal to:

$$\frac{\alpha}{\alpha - 1} x_0.$$

With a left truncation at 5 employees, we can set $x_0 = 5$, and use the method of moments to estimate α by matching a sample average among *observed firms* of 59.02 employees. We obtain a value of $\alpha = 1.0926$. Therefore, we impute the average firm size for *existing firms* by computing the expected value of a Pareto random variable with $\alpha = 1.0926$ and $x_0 = 1$. We obtain a value of 11.80 employees.

Using our estimate of the shape parameters α, we impute the dispersion in (log) firm size by acknowledging that a log Pareto distribution is equal to an exponential distribution with a rate equal to the Pareto shape. Specifically, let again the firm size be a Pareto random with scale parameter $x_0 = 1$ and shape parameter $\alpha = 1.0926$. Then the (log) of firm size follows an exponential distribution with a rate $\lambda = \alpha = 1.0926$. The standard deviation of an exponential r.v. is equal to $\sqrt{1/\lambda^2}$, implying a dispersion of (log) firm size equal to 0.915.

Finally, the overall share of innovating firms is obtained multiplying the imputed share of firms with more than 5 employees times the share of investors among *observed firms* from the WB-ES, under the assumption that firms with less than 5 employees do not engage in productivity-enhancing investment. Specifically, the probability that a Pareto random variable is larger or equal than a value x is $(x_0/x)^{\alpha}$. Using $x_0 = 1$ and $\alpha = 1.0926$, it follows that the probability of observing a firm with at least 5 employees is equal to 0.1723. Under the assumption that firms with less than 5 employees do not innovate, and given a share of *observed firms* that innovate of 0.407, the share of *existing firms* performing R&D will be equal to 0.1723 × 0.407 = 0.0701.

In the estimation, we keep the remaining three moments, i.e., the average cumulative growth rate, the average firm age, and the (log) wage dispersion, unchanged, and again apply the same data truncation to our simulated moments. Table C.7 summarizes the imputed moments used as targets, their simulated counterparts, and reports the parameter estimates.

Table C.7: Estimates and Targets - Baseline model estimated using imputed moments

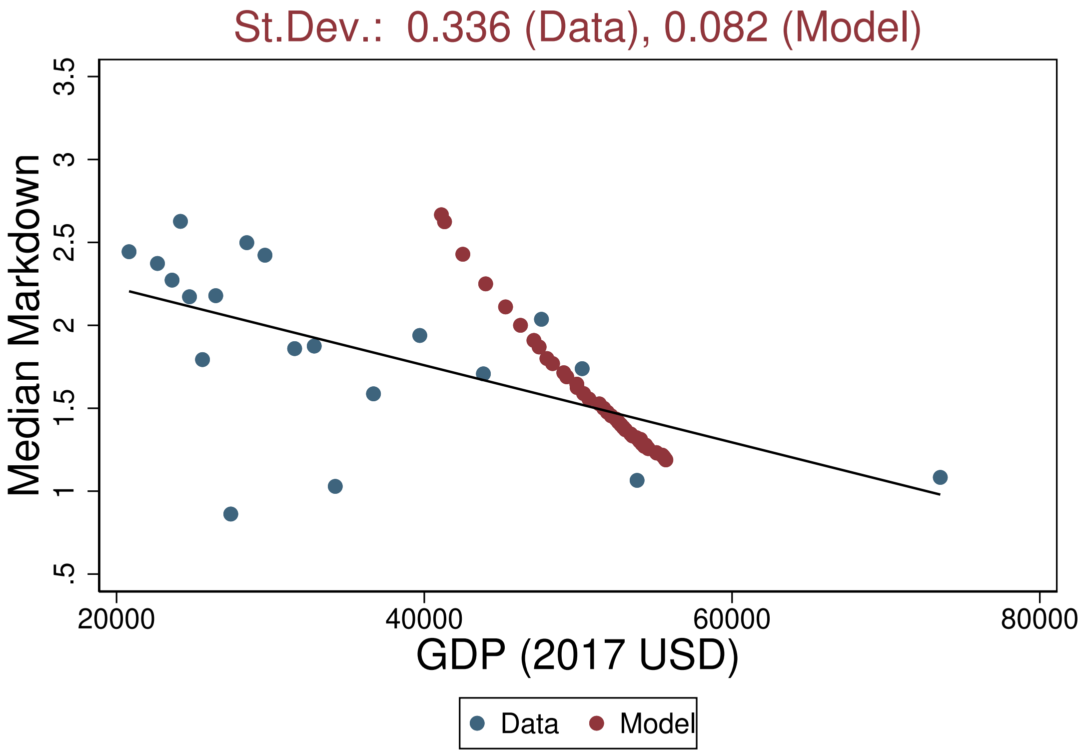
Parameters	Description	Estimates	Targets	Data	Model
			Imputed		
c_f	Operating costs	5.478	Average firm size	11.80	11.18
c_x	Innovation costs	98.34	Share of firms investing in R&D	0.070	0.066
σ_z	Productivity dispersion	2.012	Log firm size dispersion	0.915	1.064
			Observed		
p_i	Growth prob. of innovators	0.639	Average cumulative growth rate (%)	152.7	124.5
p_n	Growth prob. of innovators	0.542	Average firm age	30.13	27.78
σ_a	Amenities dispersion	0.838	Log wage dispersion	0.418	0.403

NOTES: The table shows the list of estimated parameters, their point estimates, the value of empirical targets, and model-based simulated moments. In this version of the model, we target the imputed average firm size, the imputed (log) firm-size dispersion, and the imputed share of firms investing.

Figure C.12 scatters the observed GDP per capita of each country in our sample against the estimated wage markdown (blue dots) and compares that to simulated GDP per capita obtained in counterfactual economies that feature counterfactual labor supply elasticities. As before, all other parameters are kept fixed at their benchmark values. Both measured and simulated GDP per capita are reported as fraction of the value measured for the Netherlands.

We find a standard deviation in model-based (log) GDP per capita of 0.08. It follows the model can account for about 24 percent (=0.082/0.336*100) of the observed variation in GDP per capita across countries.

Figure C.12: Cross-Country Income Differences: Model estimated using imputed moments vs. Data



NOTES: Blue dots refer to the observed GDP p.c. from panel A of Figure 5. Red dots refer to the model-based predicted GDP p.c. obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both observed and predicted GDP p.c. are expressed in USD at 2017 constant price. In this version of the model, we target the imputed average firm size, the imputed (log) firm-size dispersion, and the imputed share of firms investing. SOURCE: WDI and authors' calculation.

C.8 Estimation using alternative identification

In this section we estimate the baseline model using an alternative identification strategy. In particular, we make the model over-identified by targeting more moments than parameters. First, we replace the average firm size with the shares of firms of different size (≤ 20, 21-100, and > 100 employees). Second, we replace the average firm age with the shares of firms of different ages (≤ 30 y.o., between 30-60 y.o., and > 60 y.o.), and the overall share of firms investing in R&D with the shares of firms investing in R&D by number of employees (≤ 20, 21-100, and > 100 employees). Finally, we keep the (log) firm size dispersion and the (log) wage dispersion in the set of moments, whereas we replace the average cumulative firm size growth rate with the average annualized growth rate, computed as in Section 2, and deflated by the annual growth rate in the average firm size observed in Netherlands in the past 20 years.

Specifically, we compute the latter as follows. First, we construct the overall employment by multiplying the employment-population ratio from the World Bank Development Indicator database (variable "SL.EMP.TOTL.SP.ZS") and the working age population (variable "SL.TLF.TOTL.IN"). Hence, we construct wage and salary employment by subtracting those who are self-employed (variable "SL.EMP.SELF.ZS"). We find an annual growth rate in wage and salary employment of about 0.35 percent. Then, we use data from the Central Bureau For Statistics (CBS) on the number of active firms (retrieved from https://www.cbs.nl/en-gb/news/2023/06/enterprise-population-keeps-growing) to compute the annual growth rate in active firms. We find a value of 0.90 percent. Finally we obtain an average annual growth rate in average firm size of $0.35 - 0.90 = -0.55$ percent, and use this value to deflate the annualized growth rate in firm size obtained using the WB-ES data, i.e. 7.34 percent. We obtain and target an annualized deflated growth rate of $7.34 - (-0.55) = 7.89$ percent.

Table C.8 summarizes the parameter estimates together with empirical moments used as targets and their simulated counterparts. While there is no one-to-one mapping between parameters and targets, the identification strategy reflects the one used in the baseline estimation.

Figure C.13 relates the observed GDP per capita of each country in our sam-

Table C.8: Estimates and Targets - Overidentified baseline model

Parameters	Description	Estimates	Target	Data	Model
c_f	Operating costs	14.76	Share of firms with $\leq$ 20 employees	0.576	0.605
c_x	Innovation costs	57.96	Share of firms with 21-100 employees	0.362	0.393
p_i	Growth prob. of innovators	0.623	Share of firms with $>$ 100 employees	0.062	0.002
p_n	Growth prob. of innovators	0.546	Share of firms, $\leq$ 30 y.o.	0.615	0.685
σ_z	Productivity dispersion	2.024	Share of firms, 30-60 y.o.	0.286	0.191
σ_a	Amenities dispersion	0.778	Share of firms, $>$ 60 y.o.	0.098	0.124
			Share of firms with $\leq$ 20 employees investing in R&D	0.331	0.333
			Share of firms with 21-100 employees investing in R&D	0.504	0.457
			Share of firms with $>$ 100 employees investing in R&D	0.944	1.000
			Average annualized (deflated) growth rate (%)	6.701	7.891
			Log firm size dispersion	0.988	0.702
			Log wage dispersion	0.418	0.361

NOTES: This table reports the list of estimated parameters, their point estimates, the value of the empirical targets and their model-based simulated counterparts for the overidentified version of the baseline model.

ple (blue dots) and the counterfactual GDP per capita (red dots) against wage markdowns, keeping all other parameters fixed at their benchmark values. Both measured and simulated GDP per capita are reported as fraction of the value measured for the Netherlands. When the model is over-identified, we find a standard deviation in model-based (log) GDP per capita of 0.06. It follows the model can account for about 19 percent (=0.064/0.336*100) of the observed variation in GDP per capita across countries.

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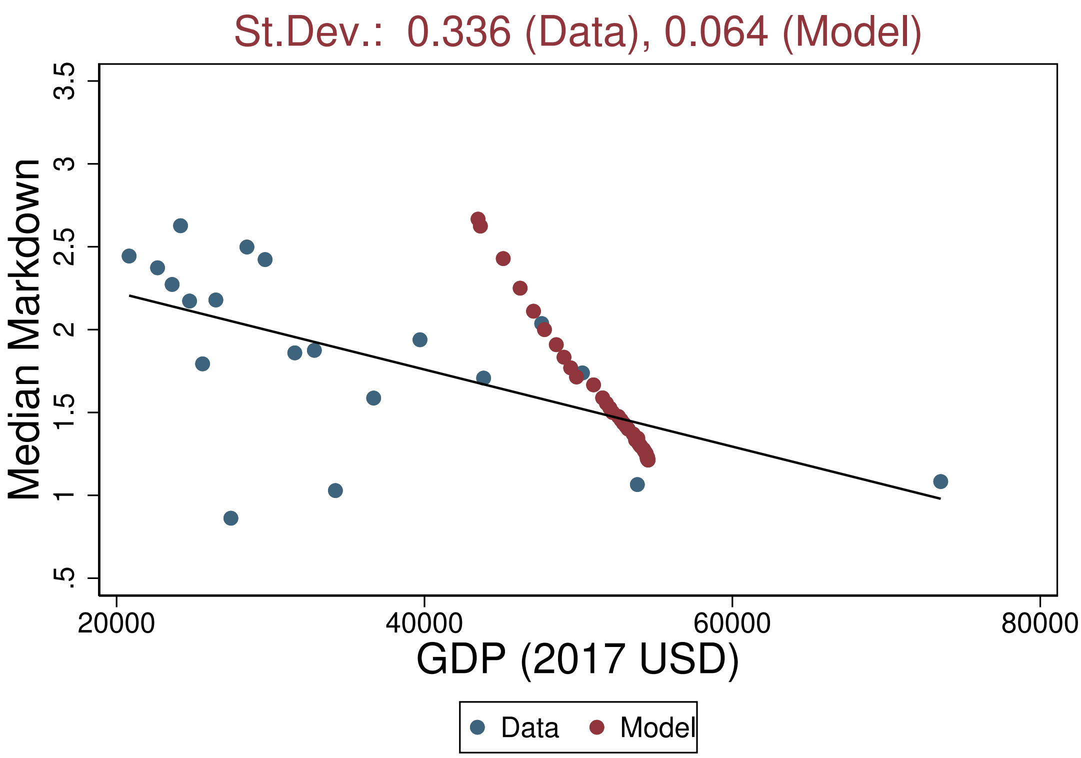
Figure C.13: Cross-Country Income Differences: Over-identified model vs. Data



NOTES: Blue dots refer to the observed GDP p.c. from panel A of Figure 5. Red dots refer to the model-based predicted GDP p.c. obtained by changing the value of ϵ^L to match the corresponding markdown in the data. Both observed and predicted GDP p.c. are expressed in USD at 2017 constant price. This version of the model is over-identified, as we target the share of firms of different sizes (# employees), the share of firms of different ages, and the share of firms investing in R&D by size, the average annualized (deflated) growth rate, (log) firm size dispersion and (log) wage dispersion. SOURCE: WDI, CBS and authors' calculation.

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